Trapped at Home: Climate Shocks, Income, and International Migration*

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Abstract: By mid-century, climate change is widely suggested to substantially increase cross-country migration, yet empirical evidence remains limited. We adopt monthly bilateral data for 127 developing origins and 180 destinations (2019-2022) and estimate migration responses to weather shocks using high-dimensional fixed effects absorbing origin-year, destination-year, and country-pair-year confounders, seasonality, and global month-specific shocks such as the COVID mobility collapse: identification rests on within-dyad seasonal variation orthogonal to the pandemic. We find no evidence that climate stress raises emigration; instead, higher temperatures reduce outflows from poor origins. Complementary income data demonstrate that weather shocks depress origin earnings. A similar response to conflict shocks suggests constrained mobility: climate change tightens origin budget constraints rather than increasing international movement.

Keywords: climate change, international migration, gravity, immobility trap, budget constraint

JEL codes: F22, J61, Q54, O15, Q15

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1 Introduction

The notion of the climate refugee has become a fixture of public debate and a focus of international policy. For example, the Institute for Economics and Peace (2020) argues that ecological threats associated with climate change put 1.2 billion people at risk of displacement by 2050. Even if only a small fraction of those affected were to move across borders,[1] this would likely be sufficient to generate large destabilising economic and political effects on the receiving countries. However, in a chapter devoted to climate-driven migration, the IPCC (2022) concludes that the evidence concerning the interactions between climate change and migration is unclear.[2] Moreover, papers directly addressing the climate change-international migration nexus are relatively scarce, especially those with worldwide coverage, and are constrained by a lack of detailed migration data for the recent time period. For instance, Beine & Parsons (2015) and Cattaneo & Peri (2016) use changes (net migration flows) in the decadal bilateral migrant stocks computed by Özden et al. (2011) over the 1960–2000 period.[3] These data limitations preclude a fixed-effect structure rich enough to absorb the many confounders potentially correlated with climate change, a difficulty that echoes long-standing concerns in the climate-income literature (Dell et al. 2014, Burke et al. 2015, Kalkuhl & Wenz 2020, Desbordes & Eberhardt 2024). We address these constraints by combining granular data on migration, climate, origin income, and domestic conflict with a battery of fixed effects tight enough to identify the effects and channels of interest under minimal omitted-variable concerns.

Our climate-migration analysis draws on the bilateral flow data of Chi et al. (2025), released through the Humanitarian Data Exchange, which aggregate Meta cross-border movement counts into a monthly dyadic panel covering 127 developing-country origins and 180 destinations over 2019–2022. The monthly frequency permits identification under a four-way fixed-effect structure: origin-destination-year, origin-by-calendar-month, destination-by-calendar-month, and year-by-calendar-month. This is richer than the origin-year, destination-year, and origin-destination design standard in the gravity literature. The origin-destination-year effect absorbs not only persistent corridor characteristics (diaspora networks, distance, shared language, bilateral labour agreements) but, being finer than an origin-year or a destination-year effect,

[1]According to Abel & Cohen (2019), the average yearly global migration flow was between 6 and 20 million, about 0.5-1% of the predicted people at risk.

[2]The IPCC writes "A general theme across studies from all regions is that climate-related migration outcomes are diverse (high confidence) and may be manifest as decreases or increases in migration flows, and may lead to changes in the timing or duration of migration and to changes in migration source locations and destinations" (p. 1079). Recent meta-analyses (Cattaneo et al. 2019, Beine & Jeusette 2021, Hoffmann et al. 2020) confirm this reading of the literature.

[3]Beine & Parsons (2015) and Gröschl & Steinwachs (2017) report no robust direct effect of climate change, whereas Cattaneo & Peri (2016) document a non-monotonic response: warmer years reduce emigration from poor origins and raise it from middle-income ones. Marchiori et al. (2012) find that weather anomalies increase international migration from Sub-Saharan African countries, while Missirian & Schlenker (2017) show that higher source-country temperature raises asylum applications to the European Union. However, both studies employ a monadic source-level specification (net international migration rate or log annual applications on source-country weather with source and year fixed effects).

every origin-year shock (origin GDP growth, exchange-rate movements, emigration or passport policy, elections and political crises) and every destination-year shock (destination labour demand and business cycle, immigration and visa-quota reforms, prevailing wages), and indeed any shock specific to the country-pair itself in a given year, such as a new bilateral labour agreement, a corridor-specific visa change, or a spell of bilateral tension. The two calendar-month effects absorb origin- and destination-specific seasonality (planting and harvest calendars, the academic year, seasonal labour demand at the destination), and the year-by-calendar-month effect absorbs global month-on-month shocks common to all dyads, including the COVID mobility collapse. This rich fixed-effect structure is precisely what insulates the estimates from the COVID shock that dominates the window: the common monthly path of the pandemic and any corridor-year COVID shock are absorbed by the fixed effects, while remaining COVID influences are captured by origin- and destination-level mobility controls. Identification of the climate elasticity rests on the residual within-corridor, within-calendar-month variation. Climate exposures are population- and cropland-weighted country-month aggregates from CRU TS v4.09 (Harris et al. 2020), with COVID controls and a twelve-month distributed lag in temperature and precipitation at both ends of the corridor.

We find that warmer or wetter conditions at either end of the corridor are associated with *less* recorded bilateral movement, not more. Climate stress keeps households at home rather than driving them abroad. At the origin, a one-degree cumulative temperature anomaly over the preceding twelve months lowers bilateral flow by 5.7 percent, while origin precipitation carries no robust effect. At the destination, both climate variables matter and both enter negatively: over the same twelve-month horizon, destination temperature reduces inflows by 5 percent and destination precipitation by 8.8 percent (per 30 mm/month). These estimates are stable across alternative fixed-effect structures, sample restrictions, different climate-aggregation weights, mobility parameterisations, a leave-one-destination test, and a placebo permutation test.

Our results support a resource-constrained 'immobility-trap' hypothesis (Black et al. 2013, Benveniste et al. 2022): a negative income shock reduces the financial ability of households to cover the cost of a cross-border move.[4] To probe this channel, we apply the same identification strategy used for the climate-migration relationship to the income response itself. A sub-national $0.5°$ grid of GDP per capita at annual frequency (Rossi-Hansberg & Zhang 2025) carries cell and country-year fixed effects that absorb persistent within-country averages and unobserved time-varying country-level confounders. A country-month nighttime-lights panel from NASA's Black Marble product (Román et al. 2018), estimated under the bilateral regression's grouped distributed-lag specification, carries country and year-month fixed effects that absorb persistent country differences and the global month-on-month path. Both panels return a precisely estimated negative temperature elasticity of origin income, while the precipitation channel is null at the origin in both the cell-level grid and the bilateral

[4] The migration-development literature supports the same interpretation: emigration rises with income in poor origins before falling at higher levels of development (Clemens 2014).

panel. The gridded estimates imply that a one-degree annual temperature anomaly reduces origin income by about 2 percent, a magnitude in line with the existing macro literature (Dell et al. 2012, Burke et al. 2015, Kalkuhl & Wenz 2020, Desbordes & Eberhardt 2024).

Lastly, we show that the negative elasticity is the signature of a destructive shock to origin livelihoods, not of climate alone. We enter a calibrated monthly conflict-risk indicator from Aizenman et al. (2026) into the bilateral regression under the same grouped distributed-lag structure as the climate variables. Origin internal conflict risk carries a negative cumulative effect on bilateral migration flows, and the headline climate coefficients are essentially unchanged once it is included. A non-climate origin shock that erodes income and the capacity to finance an orderly move thus reproduces the same contraction. The constraint mechanism is general, and the climate elasticity is not proxying for latent conflict.

The paper makes three contributions, unified by a single methodological strategy: data at three levels of resolution, each paired with a fixed-effect structure dense enough to hold the leading confounders fixed. *First*, on a monthly panel of 127 developing-country origins and 180 destinations, the bilateral climate-migration elasticity is identified under a four-way fixed-effect structure (origin-destination-year, origin-by-calendar-month, destination-by-calendar-month, year-by-calendar-month), tighter than the annual cross-country gravity literature can run on its data. The finding provides reduced-form bilateral panel evidence for the climate-immobility hypothesis that the climate-refugee discourse (Myers 1995, Brown 2008) has often overlooked. *Second*, the income channel is identified at two resolutions finer than the existing macro climate-income evidence (Burke et al. 2015, Kalkuhl & Wenz 2020): a $0.5°$ sub-national GDP grid (Rossi-Hansberg & Zhang 2025) at annual frequency with cell and country-year fixed effects, finer than the roughly 1,500-region panel of Kalkuhl & Wenz (2020); and a country-month nighttime-lights panel (Román et al. 2018) with country and year-month fixed effects, which adds temporal resolution beyond the annual cross-country standard. *Third*, we show that the constraint mechanism is not specific to climate: adding a monthly domestic conflict-risk indicator to the bilateral regression reveals that origin internal conflict risk contracts bilateral flow with the same negative sign while leaving the climate coefficients intact. A destructive non-climate origin shock thus reproduces the constraint response, a placebo test confirming that the climate elasticity captures a general erosion of the means to finance an orderly move rather than a climate-specific channel.

Section 2 sets out the framework. Section 3 describes the bilateral panel and the empirical strategy. Section 4 reports the pooled elasticities and establishes their robustness. Section 5 introduces the two income panels, tests the income channel on both, recovers the implied migration-income elasticity, and shows that a non-climate origin shock (internal conflict) reproduces the negative bilateral elasticity. Section 6 concludes.

2 A model of constrained migration

A unified household at origin i decides whether to migrate to destination j at time t. Migration is a binary choice $m \in \{0,1\}$ carrying a fixed bilateral cost c_{ij}: visa fees, recruitment fees, travel, and the financial reserves that destination immigration regimes typically demand. Household income is $y_{i,t}$; subsistence requires $\underline{c}$. The value of staying combines consumption utility over disposable income with a location amenity $A_{i,t}$ that captures the physical viability of the origin (housing, productive assets, basic infrastructure, social network):

$$V_{i,t}^{\text{stay}} \;=\; u\big(y_{i,t} - \underline{c}\big) \;+\; A_{i,t}, \tag{1}$$

with $u' > 0$, $u'' < 0$. The household solves

$$\max_{m\in\{0,1\}} \,(1-m)\,V_{i,t}^{\text{stay}} \;+\; m\,\big[V_{ij,t}^{\text{move}} - c_{ij}\big], \tag{2}$$

subject to a *financing constraint* requiring that disposable income cover the migration cost up front:

$$y_{i,t} - \underline{c} \;\geq\; c_{ij}. \tag{3}$$

Without (3) the household migrates whenever $V^{\text{move}} - c_{ij} > V^{\text{stay}}$; with it, migration additionally requires the financial slack to finance the move. The object we take to data, the bilateral flow num_{ijtm}, is the count of origin-i households executing $m = 1$ towards destination j. With households heterogeneous in income $y_{i,t}^0$ and in the cost c_{ij} they face, this count is a smooth function of a common shock even though each household's choice is discrete; the comparative statics below describe the response of that corridor-level rate, traced out by the mass of households near the two margins the model identifies.

A climate shock $\delta_{i,t} \geq 0$ enters through two channels. We write the model for a single generic stressor, but in the empirics temperature and precipitation enter as separate regressors, and the logic below applies to whichever anomaly erodes origin income or amenity. The *income channel* is a reduced-form semi-elasticity $\varepsilon_{y,T} < 0$ (the proportional change in income per degree or per precipitation increment),

$$y_{i,t}(\delta_{i,t}) \;=\; y_{i,t}^0\,\big(1 + \varepsilon_{y,T}\,\delta_{i,t}\big), \tag{4}$$

whose magnitude reflects the share of income tied to climate-sensitive activities. It is a local object: (4) linearises a relationship the macro-climate literature finds concave in the temperature level, so $\varepsilon_{y,T}$ varies across origins and steepens in hotter locations, consistent with the baseline-temperature split in Appendix A.2. The *amenity channel* is an elasticity $\varepsilon_{A,T} \leq 0$ that we leave free,

$$A_{i,t}(\delta_{i,t}) \;=\; A_{i,t}^0 + \varepsilon_{A,T}\,\delta_{i,t}. \tag{5}$$

The migration response to a climate shock operates through two margins, depending on whether the financing constraint (3) is slack or binds for a given household. When the constraint is *slack* ($y^0_{i,t} - \underline{c}$ comfortably exceeds c_{ij}), it holds with room to spare, so the move-stay decision turns purely on desirability. Differentiating the indifference condition $V^{\text{move}}_{ij,t} - c_{ij} = V^{\text{stay}}_{i,t}$ and using (4)–(5) gives the response of the flow at this *desirability margin*,

$$\left.\frac{\partial m}{\partial \delta_{i,t}}\right|_{\text{slack}} \propto -\underbrace{u'\big(y^0_{i,t} - \underline{c}\big) \cdot \varepsilon_{y,T}\, y^0_{i,t}}_{\text{income channel}} - \underbrace{\varepsilon_{A,T}}_{\text{amenity channel}} > 0. \tag{6}$$

With $\varepsilon_{y,T} < 0$ and $\varepsilon_{A,T} \leq 0$, both terms in (6) are non-negative, the income channel strictly so, leaving the right-hand side positive: V^{stay} falls, households at the desirability margin flip into migrating, and the bilateral flow rises. This is the income-push channel of the standard climate-migration narrative. When the constraint instead *binds* ($y^0_{i,t} - \underline{c} \approx c_{ij}$), a second margin is active. The same income shock pulls disposable income below c_{ij} for households sitting at the financing threshold; (3) fails, and a household that was willing and able to move is mechanically forced from $m = 1$ to $m = 0$. Its contribution to the flow is governed by

$$\left.\frac{\partial m}{\partial \delta_{i,t}}\right|_{\text{binding}} \propto \underbrace{y^0_{i,t}\, \varepsilon_{y,T}}_{\text{financing margin}} < 0, \tag{7}$$

the (resource-constrained) immobility trap: climate-driven income loss reduces realised migration not because the value of moving falls but because the migration option is mechanically removed. The two margins coexist within any corridor, and the aggregate climate response of the corridor flow is their density-weighted sum,

$$\frac{\partial m}{\partial \delta_{i,t}} = \kappa^{\text{des}}\Big[-u'\big(y^0_{i,t} - \underline{c}\big)\, \varepsilon_{y,T}\, y^0_{i,t} - \varepsilon_{A,T}\Big] + \kappa^{\text{fin}}\, y^0_{i,t}\, \varepsilon_{y,T}, \tag{8}$$

with $\kappa^{\text{des}}, \kappa^{\text{fin}} \geq 0$ the masses of households at the desirability and financing margins. The first bracket is non-negative and the second term non-positive, so the sign of the aggregate response is an empirical question, settled by which margin carries the greater mass. Slack and binding describe a household's own constraint, not a corridor: every corridor mixes households at both margins, and the sign records which margin dominates the mix. Acute amenity shocks act only through the first bracket: a large negative $\varepsilon_{A,T}$ raises it, so even where the financing margin is wide the net response can turn positive, as feasible households who would not otherwise have moved are driven out by an origin that has become physically unviable.[5]

[5]The positive source-temperature effect on asylum applications reported by Missirian & Schlenker (2017) can be read as such an amenity-dominated case. Distress migration in which *constrained* households relax the financing constraint itself (informal borrowing, distress asset sales, irregular routes) lies outside this stylised model, which reads the amenity case through the desirability margin only. Capturing it requires replacing the hard constraint (3) with a costly-financing technology in which a shortfall $\max\{0,\, c_{ij} - (y_{i,t} - \underline{c})\}$ is met at a convex penalty; that extension, derived in Appendix A.1, nests the present model and opens a second route by which the amenity channel can overturn the sign for constrained households.

The sign of the migration response is not a model prediction but an empirical question. By equation (8) it depends on the relative mass at the two margins (κ^{des} versus κ^{fin}) and on the relative magnitudes of the income and amenity elasticities, and the two signs are not equally revealing. A *negative* response can arise only when the financing margin dominates with the income channel in control: the desirability bracket in (8) is non-negative, so only its second (financing) term can pull the aggregate below zero. A *positive* response is less revealing, being consistent either with desirability-margin dominance (κ^{des} large) or, even where the financing margin carries substantial mass, with an acute amenity shock ($|\varepsilon_{A,T}|$ large) that lifts the desirability bracket above the financing term. The sign is therefore informative but not, on its own, decisive about the mechanism. Nevertheless, the financing-margin reading that a negative elasticity points to carries two auxiliary implications that an unconstrained income-push model does not. The same climate anomaly must depress origin income ($\varepsilon_{y,T} < 0$ on the same origins), and any negative non-climate origin income shock must reproduce the same negative bilateral elasticity.

The empirical strategy speaking to these theoretical considerations has three steps.

Step 1 (Sign and dominant margin). We estimate the pooled bilateral elasticity of migration to climate shocks in the dyadic migration panel (Section 4).

Step 2 (Income channel). A binding-constraint reading requires $\varepsilon_{y,T} < 0$ on the same origin countries: the temperature anomaly that contracts the bilateral flow must also depress origin-country income. We test this using a sub-national grid of GDP per capita (Section 5.1) and a country-month panel of nighttime-lights (Section 5.2).

Step 3 (Mechanism generality). If the constraint is general and not climate-specific, a negative non-climate income shock in the origin country should reproduce the same negative bilateral elasticity. We test this by adopting a calibrated monthly conflict-risk indicator (Section 5.4).

3 Climate and migration: data and empirical strategy

3.1 Bilateral migration flows

Our dependent variable is the monthly count of bilateral migration arrivals from origin i to destination j, num_{ijtm}. Flows are drawn from the Humanitarian Data Exchange release of Chi et al. (2025), which converts the privacy-protected location histories of roughly three billion monthly active Meta (Facebook) users into a bilateral migration panel covering 181 countries at monthly frequency. The construction implements the UN definition of migration through a segment-based algorithm: for each user, the algorithm identifies segments of consecutive days in which the predicted home country is the same, allowing gaps of up to 60 days. A user is counted as migrating from i to j when two adjacent segments belong to different countries, each segment lasts at least 12 months, and the user is present in the segment country on at least 50% of its days. Transit stays shorter than 12 months are not counted. The twelve-month residence requirement on each side excludes short-stay cross-border travel; the count is closer to a migration decision than to a flow of border crossings, but rare permanent moves with intermediate-length transits are missed. The same forward-residence requirement right-censors the final months of the panel: a household arriving shortly before the data end has not yet accumulated the twelve months of destination residence the definition requires, so the latest months understate the true flow. Because this censoring is common to all corridors in a given calendar month, it is absorbed by the year-by-calendar-month fixed effects of our specification (Section 3.4) and leaves the within-origin-month climate variation that identifies the climate shock effect unaffected.

A population-level estimate is then obtained by applying a *selection-rate* weighting whose denominator combines the country's Facebook penetration rate with an income-scaled adjustment. The income-scaled term receives more weight in low-income countries, where selection into Facebook usage and selection into migration both load on the same wealthier sub-population. Unfilled origin-destination-month cells are imputed as zeros, allowing extensive-margin adjustment. The sample retains origins classified as developing by the 2018 World Bank income taxonomy ($n = 127$), is unrestricted on destinations ($n = 180$), and covers the period 2019–2022 (48 monthly flow observations; climate is extended back to January 2018 so that the twelve-month distributed lag is populated from the first flow month).[6]

Figure 1 visualises the panel's corridor structure as a Sankey diagram aggregated to the seven World Bank regions: South Asia, Latin America & Caribbean, and East Asia & Pacific are the largest sending regions, and the Middle East & North Africa, Europe & Central Asia, and North America the largest recipients. Our four-way fixed-effect structure (Section 3.4) should absorb any persistent corridor-level discrepancy between the population-representative estimate and the unobserved true flow.[7]

3.2 Climate exposures

Monthly country-level temperature ($\textsc{tmp}$) and precipitation ($\textsc{pre}$) are aggregated from the CRU TS v4.09 0.5° gridded series as

$$\bar{x}_{i,t,m} = \frac{\sum_{c \in i} w_c \, A_{c,i} \, x_{c,t,m}}{\sum_{c \in i} w_c \, A_{c,i}}, \tag{9}$$

where c indexes cells inside country i, $A_{c,i}$ is the fractional area of cell c inside country i computed from the GADM boundaries, and w_c is a cell weight. The baseline uses population

[6]Twelve developing economies have no Meta-recorded outflow and do not enter the panel as origins; the populous cases (China, Iran, North Korea, and Cuba) are countries where Meta platforms are blocked or restricted, so their absence is geopolitical rather than evidence of zero emigration; the remainder are small island micro states or disputed territories. The estimation sample is therefore confined to developing origins where Facebook coverage is sufficient to observe migration.

[7]In addition, we compare our data against Eurostat's annual bilateral immigration counts for EU-27 destinations over the same 2019–2022 window. The two series correlate at 0.89 in logs across 10,257 corridor-year observations.

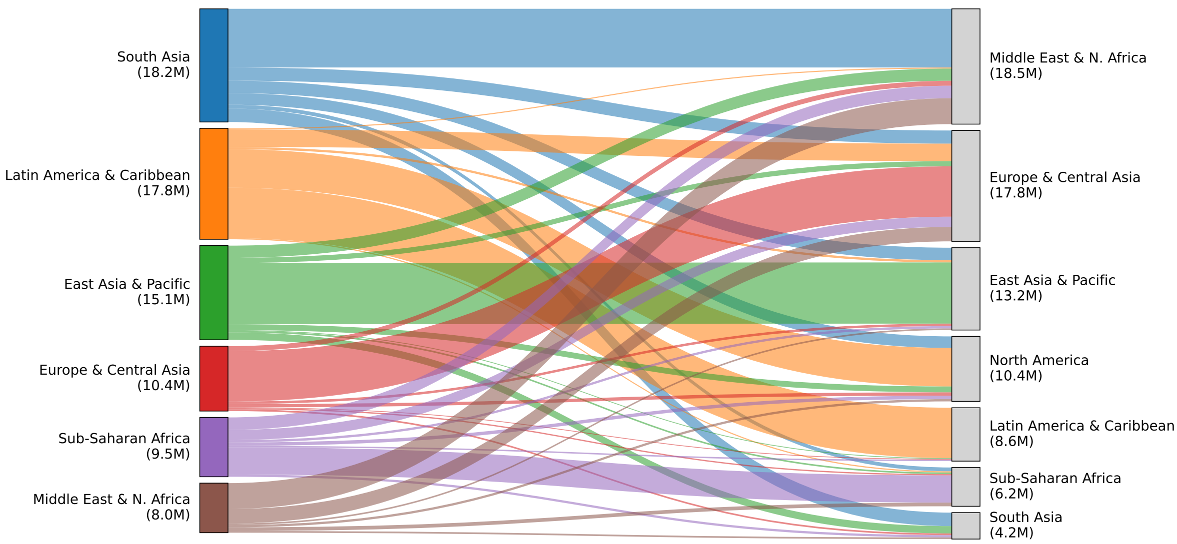


Figure 1: Migration flows by World Bank region, 2019-2022. Sending region on the left, receiving region on the right; ribbon width is proportional to total bilateral migration count over the panel window in millions of migrants. Origins restricted to the developing-country estimation sample; destinations unrestricted. Region-by-region corridors with total flow below 100,000 migrants omitted.

weights: migration is initiated by people, and the relevant exposure is the climate the average resident faces. Precipitation is divided by 30 mm before estimation so that one unit corresponds to roughly the same share of within-origin-month variability as one degree Celsius of temperature, putting the temperature and precipitation coefficients on comparable magnitudes.[8]

We construct five weighting schemes, each fixed at the 2015 vintage to keep the weight exogenous to the 2019–2022 climate variation captured in the regressions. The *area* weight uses $\cos(\phi_c)$ at cell latitude ϕ_c. The *population* weight uses Gridded Population of the World v4 rev11 (Center for International Earth Science Information Network (CIESIN), Columbia University 2018) cell counts at 0.5°, adjusted to UN World Population Prospects 2015 country totals. The *cropland* weight uses GAEZ+ 2015 monthly cropland data (Grogan et al. 2021). The *GDP* weight uses cell-level gridded GDP from the BFI Data Studio sub-national release (Rossi-Hansberg & Zhang 2025). The *low-income* weight uses $\text{pop}_c^2/\text{GDP}_c$, equivalent to population weighted by inverse GDP per capita; both numerator and denominator are from BFI. The five weights span the within-country distribution of who bears the climate exposure: area weights the country as a unit of land, population the average resident, GDP the average dollar of output, low-income the population weighted by inverse GDP per capita, and cropland the average farmer. Section 4.2 exploits the differences. The 2019–2022 temperature and precipitation anomalies that drive our estimates are, by historical standards, ordinary: their distribution is statistically indistinguishable from a 1990–2018 baseline (Appendix A.5), so the

[8]The rescaling only changes the absolute scale of the reported precipitation elasticity (multiplying any reported precipitation coefficient by 30 recovers the mm-per-month interpretation) and does not affect statistical significance.

short panel captures normal weather variation rather than an anomalous window.

3.3 Controls and corridor classifiers

We control for COVID mobility restrictions using the Oxford COVID-19 Government Response Tracker (OxCGRT). $\mathrm{mob}_{i,tm}^{\mathrm{COVID}}$ and $\mathrm{mob}_{j,tm}^{\mathrm{COVID}}$ are the first principal component of four OxCGRT sub-indicators (public transport closure, stay-at-home requirements, internal-movement restrictions, international-travel controls) at origin and destination, entered contemporaneously and with a one-month lag. The PCA score is normalised to mean zero and unit variance over 2020–2022 and set to zero in 2018–2019.

Three corridor-quality flags (Gulf labour, circular, transit) enter the sample-restriction robustness of Section 4.2.1, intended to gauge the robustness of our the estimates against anomalous/exceptional migration patterns; their definitions are provided in Table 1, which reports the rule used to construct each flag and the share of observations it marks. Table 2 reports summary statistics for the estimation panel.

Table 1: Definitions of corridor-quality flags used in the sample-restriction robustness.

Flag	Definition (corridor $i \to j$; panel totals over 2019–2022)	Share of observations
Gulf	Destination is a Gulf Cooperation Council state (Saudi Arabia, United Arab Emirates, Qatar, Kuwait, Bahrain, Oman).	3.4%
Transit	Destination is one of thirteen documented transit hubs: Belarus, Colombia, Ecuador, Greece, Guatemala, Honduras, Libya, Morocco, Mexico, Nicaragua, Panama, Serbia, Türkiye.	7.2%
Circular	Two-way corridor: the smaller directional flow exceeds half the larger ($\min / \max > 0.5$) and the combined two-way flow falls in the top decile across all corridors.	4.2%
Any	Any of the three above; the clean sample is its complement (86% of observations).	14.0%

Table 2: Descriptive statistics, monthly bilateral migration panel, 2019–2022.

	Mean	SD	p10	Median	p90	Min	Max
Bilateral migration count (thousands)	0.08	1.03	0.00	0.00	0.05	0.00	163.94
Origin temperature (°C)	21.87	7.46	11.58	24.02	28.37	-21.93	36.94
Origin precipitation (mm/day)	3.49	3.64	0.15	2.23	8.52	0.00	34.90
Destination temperature (°C)	19.88	8.72	6.22	22.58	28.23	-21.93	39.41
Destination precipitation (mm/day)	3.21	3.34	0.15	2.16	7.82	0.00	34.90
Origin COVID mobility index	0.04	1.73	-1.39	-0.93	2.87	-1.39	5.00
Destination COVID mobility index	-0.04	1.66	-1.39	-1.01	2.70	-1.39	5.00

3.4 Specification

The estimation equation for bilateral flows is

$$\begin{aligned}
\mathsf{E}[\mathsf{num}_{ijtm} \mid \cdot] \;=\; \exp\Big\{ &\sum_{\ell=0}^{L} \big(\beta^o_\ell \mathrm{TMP}_{i,tm-\ell} + \gamma^o_\ell \mathrm{PRE}_{i,tm-\ell} + \beta^d_\ell \mathrm{TMP}_{j,tm-\ell} + \gamma^d_\ell \mathrm{PRE}_{j,tm-\ell}\big) \\
&+ \sum_{\ell=0}^{1} \big(\eta^o_\ell\, \mathsf{mob}^{\mathsf{COVID}}_{i,tm-\ell} + \eta^d_\ell\, \mathsf{mob}^{\mathsf{COVID}}_{j,tm-\ell}\big) \\
&+ \phi \ln(\mathsf{num}_{ij,tm-1} + 1) \;+\; \psi\, \mathbf{1}[\mathsf{num}_{ij,tm-1} = 0] \\
&+ \alpha_{ijt} + \alpha_{im} + \alpha_{jm} + \alpha_{tm} \Big\},
\end{aligned} \tag{10}$$

where TMP and PRE are the origin's (superscript o) and destination's (d) population-weighted temperature and precipitation in month m lagged up to $L = 12$ months; the COVID controls, lagged dependent variable, and extensive-margin indicator in rows 3 and 4 are detailed below; and $\alpha_{ijt}, \alpha_{im}, \alpha_{jm}, \alpha_{tm}$ are origin-destination-year, origin-calendar-month, destination-calendar-month, and year-calendar-month fixed effects. We estimate this model by Poisson Pseudo-Maximum-Likelihood with high-dimensional fixed effects (Correia et al. 2020), standard for gravity equations with positive and zero flows (Santos Silva & Tenreyro 2006).

The four fixed-effect sets absorb the following. The origin-destination-year effect α_{ijt} removes everything constant within a corridor in a given year. Because it is finer than a simple origin-year or destination-year effect, it absorbs not only persistent dyadic migration intensity (diaspora networks, distance, shared language, colonial ties, bilateral labour agreements) but also any *origin-year* shock (origin GDP growth, exchange-rate movements, national emigration or passport policy, elections and political crises) and any *destination-year* shock (destination labour demand and business cycle, immigration and visa-quota reforms, prevailing wages, asylum-regime changes). The origin-calendar-month effect α_{im} removes origin-specific seasonal cycles (planting and harvest calendars, the academic year, religious-calendar timing such as Ramadan and the Hajj). The destination-calendar-month effect α_{jm} removes destination-specific seasonal labour demand (agricultural, construction, and tourism seasons). The year-calendar-month effect α_{tm} removes global month-on-month shocks common to all dyads, including the COVID mobility collapse, world commodity prices, and the global business cycle. The variation remaining in climate shocks is the deviation of country i's temperature in month tm from its origin-by-month seasonal norm, purged of the year-by-month global path. Identification of β^o_ℓ rests on this within-origin-month residual variation, distributed over the 12-month lag. Similarly for δ^o_ℓ and the equivalent destination terms. Although the panel spans only four years, this identifying variation is not peculiar to the window: its distribution over 2019–2022 is statistically indistinguishable from a 1990–2018 baseline, so the elasticity is not an artifact of an anomalous climatology (Appendix A.5). The maintained identifying assumption is that, conditional on the four-way fixed effects, no within-origin-month determinant of bilateral flow co-moves with the origin climate anomaly; the temporal-permutation test of Section 4.2.5 probes, but cannot by itself prove, this assumption. Standard errors are clustered at the origin-destination dyad. The cumulative effect at horizon h reported in subsequent tables is the sum of the grouped-lag coefficients over $\ell = 0, \dots, h$, with groups contemporaneous ($\ell = 0$), short-lag ($\ell = 1, \dots, 4$), medium-lag ($\ell = 5, \dots, 8$), and long-lag ($\ell = 9, \dots, 12$). Grouping is less susceptible to inter-lag multicollinearity than the individually-unconstrained alternative.

Lagged-flow controls. The bilateral flow $\text{num}_{ij,tm}$ combines an intensive margin (how many migrants move through an already-active migration corridor) and an extensive margin (whether the corridor is active at all). Roughly one-third of dyad-months in our panel have $\text{num}_{ij,tm} = 0$ (Table 2). The four-way fixed-effect structure pins down a corridor-year average level of activity and the calendar-month and global timing baselines, but does not absorb within-corridor month-to-month dynamics in either margin. The lagged dependent variable $\ln(\text{num}_{ij,tm-1} + 1)$ enters with coefficient ϕ and identifies intensive-margin persistence on active dyads. The extensive-margin indicator $\mathbf{1}[\text{num}_{ij,tm-1} = 0]$ enters with coefficient ψ and captures the discrete shift for dyads that were dormant in $tm - 1$; without it, ϕ would absorb both the smooth persistence on active corridors and the kink at $\text{num}_{ij,tm-1} = 0$. Together the two controls prevent the climate elasticity from absorbing persistence that is itself auto-correlated with monthly climate. The Nickell-type bias on ϕ is of order $1/T$ in our 48-month panel, and any contamination of the climate coefficient through ϕ is bounded by ϕ times the small climate-LDV covariance. Because the lagged dependent variable enters the specification, the cumulative climate effects reported below are conditional on lagged flow and are best read as short-run, conservative estimates of the total dynamic response; we do not rescale them to a long-run elasticity by a $1/(1 - \phi)$ multiplier, since the origin-destination-year fixed effects absorb each corridor's annual mean flow and a long-run steady-state level is not identified.

4 Climate and migration: the pooled bilateral elasticity

4.1 The pooled elasticity

We estimate the grouped-lag specification (10) of bilateral migration on origin and destination temperature and precipitation, pooled over the developing-origin sample.

The upper panel of Figure 2 plots the cumulative effect of origin temperature and precipitation on bilateral flow as the distributed-lag window widens from the contemporaneous month out to twelve months, each lag entered without constraint. The origin-temperature profile descends into negative territory over the first months, settles at its mature value by about the eighth month, and is flat thereafter, the timing expected for transmission through a growing-season-length window. The origin-precipitation profile stays close to zero throughout.

Table 3 reports the cumulative grouped-lag coefficients at the 8- and 12-month horizons. The coefficient on origin temperature is negative and statistically significant at both horizons.

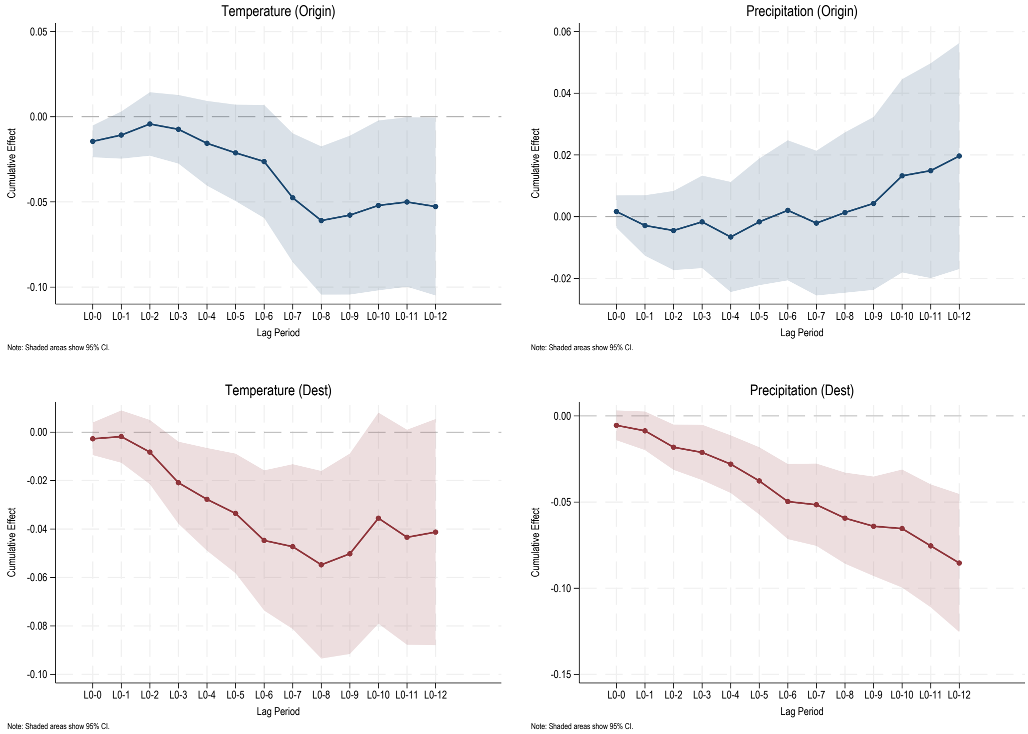


Figure 2: Cumulative individual-lag effects of origin temperature and precipitation on bilateral migration flows. Lags $\ell = 0, \ldots, 12$ are individually unconstrained.

Table 3: Pooled bilateral elasticity of migration to climate: cumulative grouped-lag effects of origin and destination temperature and precipitation.

Climate grouped-lag	8-month	12-month
Origin temperature	-0.063^{***}	-0.057^{**}
	(0.022)	(0.027)
Origin precipitation	$+0.002$	$+0.019$
	(0.013)	(0.018)
Destination temperature	-0.064^{***}	-0.051^{**}
	(0.020)	(0.023)
Destination precipitation	-0.062^{***}	-0.088^{***}
	(0.013)	(0.020)

Notes: Poisson PML with high-dimensional fixed effects on the developing-origin bilateral panel, 919,532 observations. Each climate variable enters under the grouped distributed-lag windows $\{0, 1\text{-}4, 5\text{-}8, 9\text{-}12\}$; cells are cumulative effects at the 8- and 12-month horizons (temperature per degree Celsius, precipitation per 30 mm/month). Four-way fixed effects (origin-destination-year, origin-month, destination-month, year-month); cluster-robust standard errors at the origin-destination dyad in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Read as an elasticity,[9] the 8-month cumulative effect implies that a one-degree cumulative temperature anomaly over the preceding eight months lowers bilateral flow by about 6.3 percent, and the 12-month effect by 5.7 percent. The coefficient on origin precipitation is small and not statistically distinguishable from zero at either horizon.

The equivalent plots for destinations in the lower panel of the same Figure are also downward-sloping. In Table 3, the coefficient on destination temperature is negative and statistically significant, of similar magnitude to the origin one, with a one-degree anomaly lowering inflows by about 6.4 percent at eight months. The coefficient on destination precipitation is negative and statistically significant, implying that a 30 mm/month anomaly lowers inflows by about 6.2 percent at eight months and 8.8 percent at twelve. The precipitation response is thus asymmetric, operating at the receiving end of the migration corridor but not at the sending end; the destination coefficients are consistent with a labour-demand channel in the sectors that hire the marginal migrant, and the eight- to twelve-month accumulation rules out a purely mechanical short-run transport disruption. Our specification identifies the reduced-form destination effect but cannot separate the candidate channels behind it (labour demand, amenity, or administrative processing); doing so would require destination-side labour-market data that the present panel lacks.[10]

4.2 Robustness

The headline origin temperature coefficient is stable under alternative fixed-effect structures, sample restrictions, climate-aggregation weights, COVID parameterisations, and a leave-one-destination test.[11]

4.2.1 Alternative fixed effects and corridor restrictions

Table 4 reports the cumulative 8- and 12-month coefficients on the four headline climate variables under six alternative specifications. Replacing the additive origin-by-month and destination-by-month fixed effects with a single corridor-by-month fixed effect α_{ijm} in column (2) leaves the origin-temperature 8-month coefficient at -0.066, indistinguishable from the baseline; the result does not merely reflect corridor-specific seasonal patterns that the additive

[9] We use 'elasticity' throughout as shorthand for the response of log bilateral flow to a one-unit climate change (one degree Celsius for temperature, 30 mm/month for precipitation). Because the climate variables enter in levels, this is strictly a semi-elasticity. The implied migration-income object of Section 5.3, $\varepsilon_{m,y}$, is by contrast a true (log-log) elasticity.

[10] From here on we focus mainly on origin climate shocks, and especially on high origin-temperature anomalies, the variation that carries the income channel at the heart of the constraint mechanism; the destination and origin-precipitation responses are not pursued further.

[11] It also shows no income gradient across origins (Appendix A.2): the pooled elasticity is representative of the developing-origin income distribution. Appendix A.2 additionally documents a pronounced gradient in baseline temperature: the negative response is concentrated in already-hot origins. Nor is the response amplified by pre-existing migrant networks: a literature on US-bound corridors finds that a larger diaspora can convert a climate shock into a positive emigration push (Mahajan & Yang 2020, Ibáñez et al. 2026), but interacting origin temperature with the 2015 log bilateral migrant stock yields a negative, not positive, interaction, so the trapping response is if anything stronger in high-network corridors (Appendix A.3).

structure fails to absorb. Interacting every climate grouped-lag with the binary flag combining the Gulf, circular, and transit indicators in (3a) yields a clean-corridor (86 percent of all observations) elasticity of -0.075 and a *differential* interaction on the flagged 14 percent of $+0.022$, reported in (3b), that is not statistically significant; the implied flagged-corridor elasticity is -0.053, indistinguishable from the clean-corridor one. Dropping the ten dyads with the largest total bilateral flow (Gulf labour pipelines, MEX-USA, VEN-COL, UKR-POL, among others) in (4) yields an 8-month origin-temperature elasticity of -0.054: modestly attenuated, still significant. The climate-flow elasticity is not a property of Gulf or circular corridors and is not driven by a small number of large dyads.

Columns (5) and (6) address spatial autocorrelation in climate shocks, which would otherwise inflate the precision of the dyad-clustered baseline. Weather systems are spatially extended, so residuals across observations whose origins share weather systems may be correlated even after the four-way fixed-effect structure. Column (5) re-clusters the baseline regression three ways: on the dyad and on the M49 sub-region $\times$ year-month at both ends of the corridor.[12] Column (6) augments the baseline regression with row-standardised inverse-distance neighbour-weighted climate (within 2,000 km) as additional regressor (a spatial-lag-of-X, or SLX, specification), so the reported coefficient identifies the own-country effect holding the neighbour-spillover constant. The 8-month origin-temperature elasticity survives both robustness checks at the 5% level; the destination precipitation channel survives at the 1% level in every column.

4.2.2 Weighting schemes

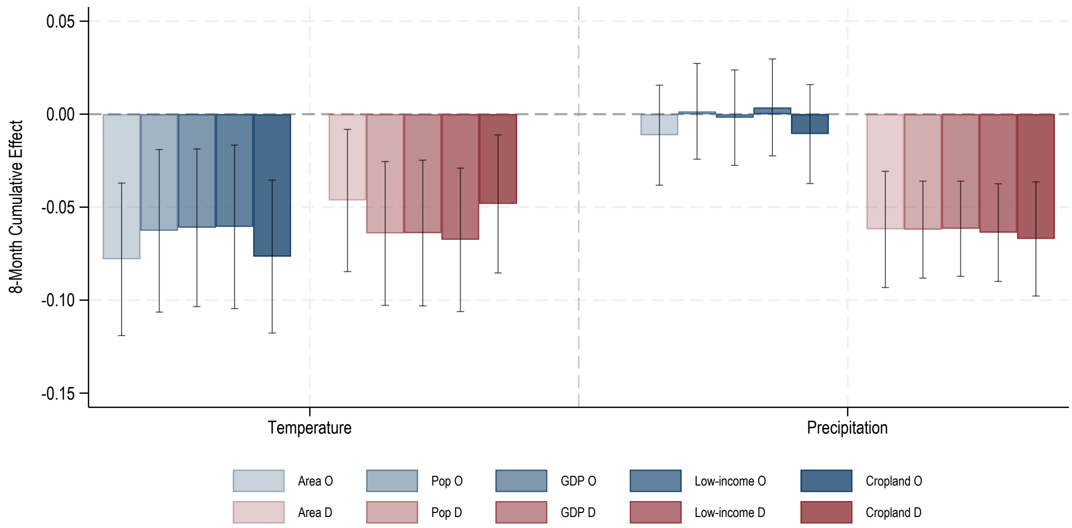
Figure 3 compares the 8- and 12-month cumulative climate shock coefficients under the five weighting schemes of Section 3.2. These weights load on different subsets of the country's spatial distribution: area weights on the average square kilometre, population on the average resident, GDP on the average dollar, low-income on the population weighted by inverse GDP per capita, cropland on the average farmer. The estimates are stable across all five schemes because, despite within-country cell-level population, GDP, and cropland correlating only at around 0.4, climate shocks tend to span the full country territory, so the weighting choice barely moves the country-month aggregate. The weights matter at all only in the seven geographically largest developing countries (Brazil, Mexico, Indonesia, China, India, Russia, the United States), where the population, GDP, and cropland weights pick out different regions of a vast territory, across which a temperature anomaly need not be uniform.

[12] M49 is the United Nations Statistics Division geographic classification (*Standard Country or Area Codes for Statistical Use*, Series M, No. 49), which partitions the world's countries into 22 sub-regions (e.g. Western Africa, South-Eastern Asia, Northern Europe). Re-clustering at the M49 sub-region $\times$ year-month allows residuals to be correlated across all corridors whose origin (or destination) sits in the same sub-region in the same month, the relevant unit of common weather exposure for a synoptic-scale shock.

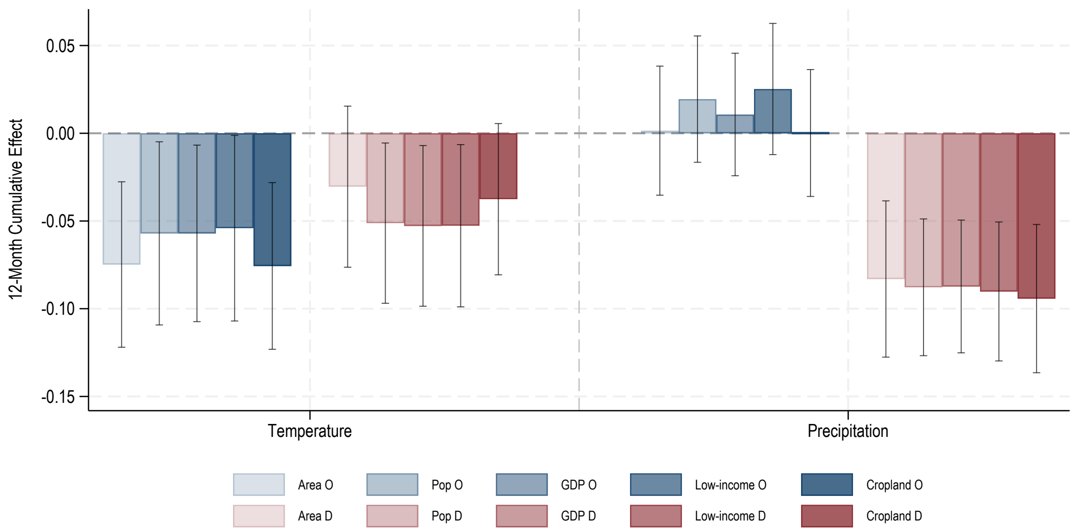
Table 4: Headline cumulative effects

	(1) Baseline	(2) Corridor-month FE	(3a) Clean main effect	(3b) Differential flagged	(4) Drop top-10 dyads	(5) Three-way cluster (M49)	(6) SLX 2000km dyad cluster
Origin temperature							
8m	-0.063^{***}	-0.066^{***}	-0.075^{***}	$+0.022$	-0.054^{**}	-0.063^{**}	-0.073^{**}
	(0.022)	(0.022)	(0.026)	(0.033)	(0.022)	(0.030)	(0.035)
12m	-0.057^{**}	-0.064^{**}	-0.081^{**}	$+0.045$	-0.046^{*}	-0.057	-0.085^{*}
	(0.027)	(0.026)	(0.032)	(0.047)	(0.026)	(0.036)	(0.045)
Origin precipitation (per 30mm)							
8m	$+0.002$	$+0.002$	$+0.008$	-0.014	-0.016	$+0.002$	$+0.012$
	(0.013)	(0.013)	(0.016)	(0.022)	(0.013)	(0.016)	(0.013)
12m	$+0.019$	$+0.018$	$+0.028$	-0.016	-0.005	$+0.019$	$+0.030$
	(0.018)	(0.018)	(0.023)	(0.033)	(0.019)	(0.021)	(0.018)
Destination temperature							
8m	-0.064^{***}	-0.057^{***}	-0.079^{***}	$+0.031$	-0.042^{**}	-0.064^{**}	-0.063^{***}
	(0.020)	(0.020)	(0.021)	(0.030)	(0.019)	(0.032)	(0.018)
12m	-0.051^{**}	-0.042^{*}	-0.065^{**}	$+0.029$	-0.028	-0.051	-0.052^{**}
	(0.023)	(0.024)	(0.027)	(0.042)	(0.023)	(0.040)	(0.022)
Destination precipitation (per 30mm)							
8m	-0.062^{***}	-0.061^{***}	-0.050^{***}	-0.024	-0.064^{***}	-0.062^{***}	-0.058^{***}
	(0.013)	(0.013)	(0.019)	(0.026)	(0.015)	(0.017)	(0.013)
12m	-0.088^{***}	-0.089^{***}	-0.069^{**}	-0.035	-0.083^{***}	-0.088^{***}	-0.085^{***}
	(0.020)	(0.020)	(0.030)	(0.037)	(0.021)	(0.024)	(0.020)
N	919,532	830,990	919,532		919,062	919,532	919,532

Notes: Poisson PML with high-dimensional fixed effects on the bilateral panel. Cumulative effects are sums of grouped-lag coefficients. Standard errors in parentheses. Spec (1) baseline (dyad cluster); Spec (2) replaces $\alpha_{im} + \alpha_{jm}$ with α_{ijm}; Spec (3) interacts every climate grouped-lag with the binary corridor-quality flag (Gulf, circular, or transit), column (3a) the main effect (clean corridors), column (3b) the differential on the flagged sub-sample (flagged-corridor elasticity = (3a)+(3b)); Spec (4) drops the ten dyads with the largest total bilateral flow. Spec (5) re-clusters the baseline three-way at the dyad and at the M49 sub-region × year-month at both ends of the corridor. Spec (6) augments the baseline regression with row-standardised inverse-distance neighbour-weighted climate (2000 km cutoff) constructed inline from CEPII bilateral distances; reported coefficients are the own-country effects holding the neighbour spillover constant, dyad-clustered. Origin and destination COVID mobility indices, lagged $\ln$(flow), and lagged zero-flow indicator included throughout. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) 8-month cumulative effect



(b) 12-month cumulative effect

Figure 3: Climate effects under five country-level weighting schemes (area, population, GDP, low-income, cropland) at the 8-month (top) and 12-month (bottom) cumulative horizon. The low-income weight is pop^2/GDP from BFI 2015, equivalent to population weighted by inverse GDP per capita. The GDP and low-income weights share a common sample (BFI 2015 coverage); the cropland-weighted regression drops country-months with zero cropland coverage (very high-latitude or hyper-arid months), which slightly reduces its sample relative to the area- and population-weighted regressions. 95 percent confidence intervals.

4.2.3 COVID specification

The 2020–2021 portion of the panel coincides with the steepest peacetime mobility restrictions on record. We test for confounding with the COVID mobility collapse in two ways. First, we vary the functional form of the COVID controls: the baseline contemporaneous-and-lagged linear form is replaced by year-varying linear interactions, by a quadratic in the mobility index, and by the four sub-indicators entered separately. Second, we drop the COVID controls entirely and rely on the four-way fixed-effect structure alone. Figure 4 reports the 8- and 12-month cumulative climate coefficients across all four specifications. The estimates are visually indistinguishable: the headline negative origin-temperature elasticity, the destination-temperature effect, and the destination-precipitation effect retain their sign, magnitude, and significance in every parameterisation.

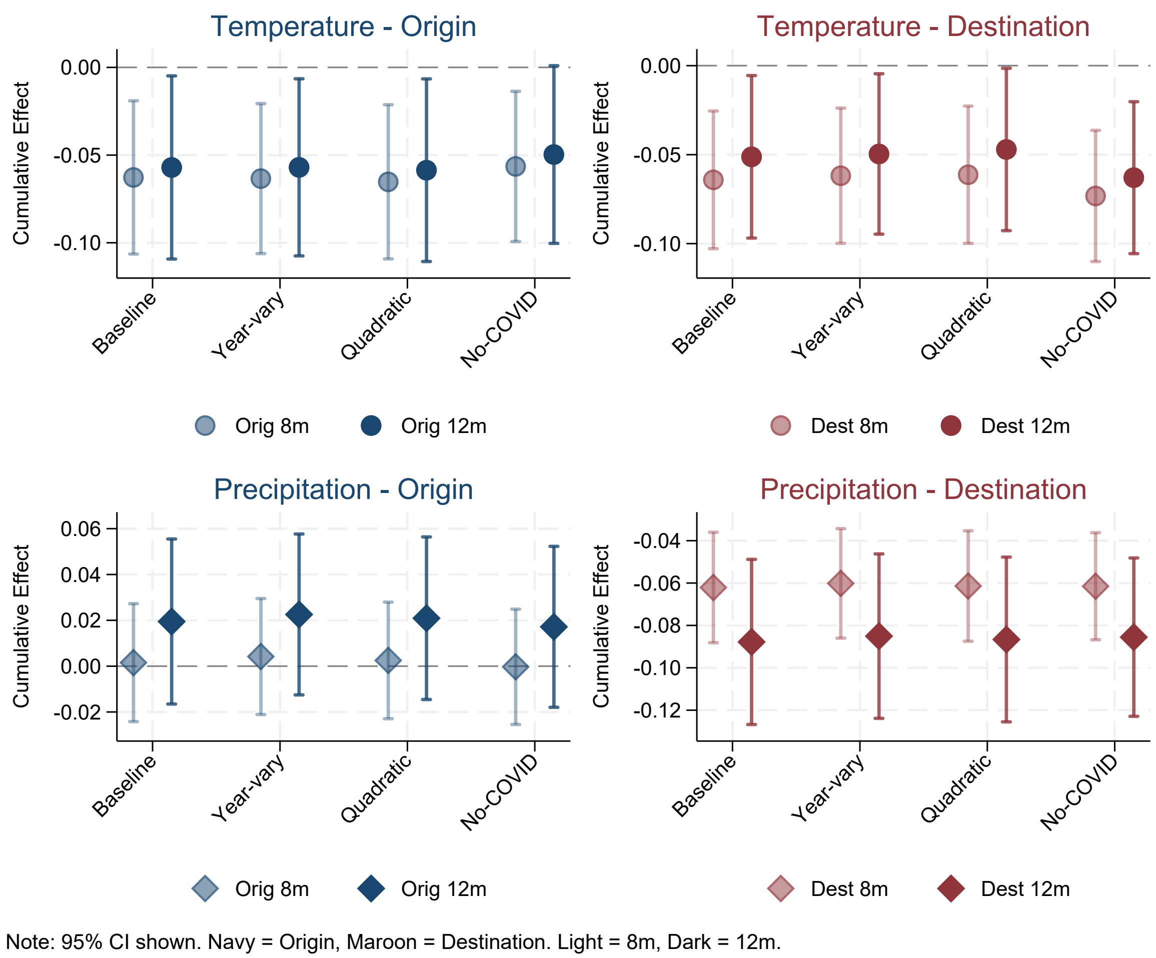


Figure 4: Robustness to COVID specification: 8-month and 12-month cumulative climate effects under alternative COVID parameterisations.

4.2.4 Leave-one-destination-out

We re-estimate the baseline 20 times, each time dropping one of the 20 destinations with the largest cumulative inflows from developing origins. Figure 5 reports the 8-month origin-temperature elasticity in each leave-out specification. The estimates cluster tightly around the full-sample point, ranging over $[-0.053, -0.069]$, with every 95 percent confidence interval excluding zero. No single destination moves the elasticity into a qualitatively different range, whether a Gulf state, the United States, or a large European destination.

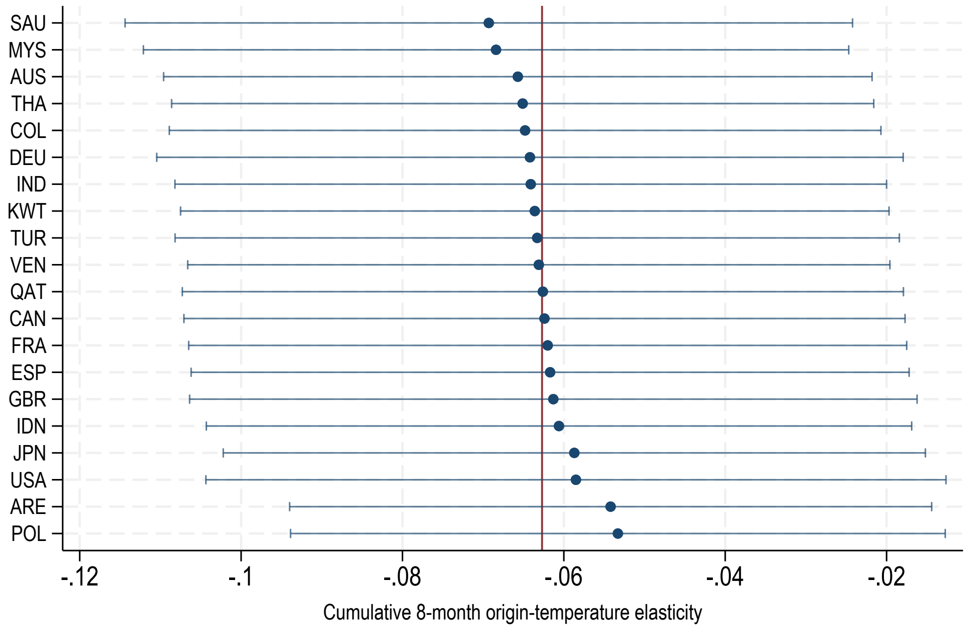


Figure 5: Leave-one-destination-out robustness, 8-month cumulative origin-temperature elasticity. Each row drops one of the top-20 destinations by total flows and re-estimates equation (10) on the remainder. Whiskers: 95 percent CI. Vertical reference line: full-sample point estimate (-0.063).

4.2.5 Temporal permutation: a placebo test of causality

We implement a permutation exercise as a placebo test of causality: it asks whether the estimated climate-migration relationship requires the actual timing of climate shocks, or whether a similar coefficient would arise from reshuffled climate that preserves only each origin's climatology. We randomly reassign each origin's observed monthly temperature and precipitation across months, independently for each origin, preserving each origin's own climate distribution but scrambling the calendar timing, which destroys the alignment between a climate anomaly and the bilateral flow in the same month. The reshuffling severs the hypothesised causal channel while leaving each origin's marginal climate variation intact. If the estimated effect reflects the causal effect of the timing of climate shocks on migration, scrambling that timing should drive the coefficient toward zero; if it were instead a spurious feature of the panel's structure, it would survive the reshuffling.

For each origin we shuffle the 60 calendar-month observations of temperature and precipitation 100 times, preserving the origin's identity but breaking the alignment between climate at month tm and flow at month tm. Shuffles are independent across origins. The same procedure is applied independently at the destination side. We re-estimate (10) on each shuffled panel and store the cumulative 8- and 12-month coefficients.

Figure 6 plots the placebo distributions. The observed origin- and destination-temperature coefficients at both horizons, and destination precipitation, fall well outside their placebo distributions: the effects are present only when the true climate–flow timing is preserved and vanish once it is scrambled, which is the pattern expected of a causal response, and not of a timing-independent artifact such as seasonality or the panel's finite length. Origin precipitation lies inside its placebo distribution, mirroring the headline origin-precipitation null. Spatial autocorrelation is addressed separately, and conservatively, by the sub-region re-clustering and the inverse-distance spatial-lag control of Section 4.2.1. Procedural details are provided in Appendix A.4.

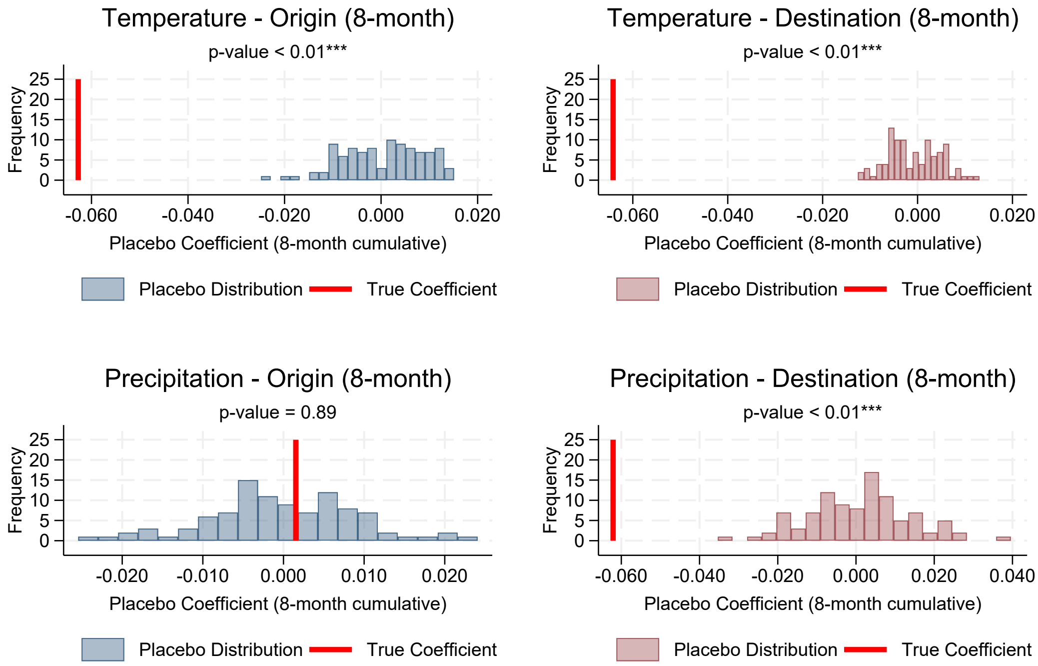
5 The constraint mechanism: income and a non-climate shock

Section 4 delivered Step 1 of our empirical strategy: the pooled bilateral elasticity is negative, so the financing margin dominates the typical developing-origin corridor, with the income channel in control. We now execute Steps 2 and 3. Step 2 tests the income channel on two income panels for the same origin countries at different temporal and spatial resolutions: an annual cell-level GDP per capita panel from Rossi-Hansberg & Zhang (2025) (Section 5.1) and a country-month nighttime-lights panel we build from the NASA Black Marble VNP46A2 daily product (Section 5.2). The two panels measure the same income response at the annual within-cell and the monthly within-country-year margins. Section 5.3 combines the cell-level elasticity with the bilateral elasticity through (13) to recover an implied migration-income elasticity. Step 3, in Section 5.4, shows that a negative non-climate origin income shock (internal conflict) reproduces the same negative bilateral elasticity.

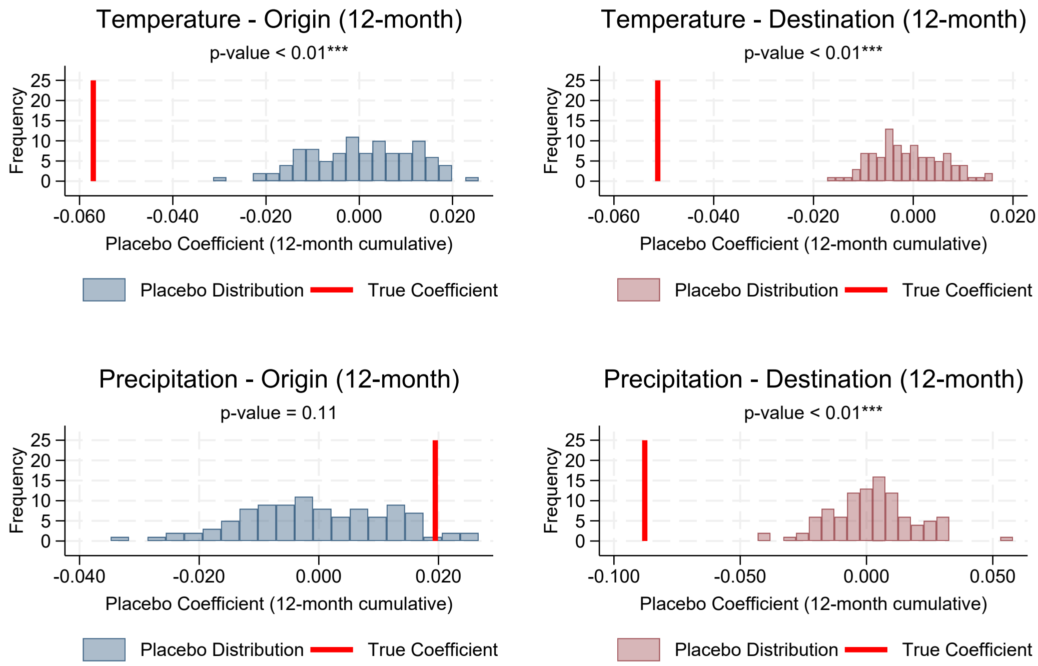
5.1 The income response at the grid level

The constraint mechanism rests on a premise the bilateral panel cannot test directly: temperature anomalies at the origin reduce household income. We test this on a sub-national grid of GDP per capita built for the same developing-country origins.

The grid data come from the BFI Data Studio sub-national GDP-per-capita panel (Rossi-Hansberg & Zhang 2025), which assigns annual GDP per capita to roughly 73,000 0.5° land cells through a random-forest predictor trained on county- and state-level GDP for the subset of countries that report such data. The predictor inputs are unsaturated nighttime-light radiance from the VIIRS Black Marble product (building on Henderson et al. 2012), LandScan gridded



(a) 8-month cumulative effect



(b) 12-month cumulative effect

Figure 6: Temporal permutation distribution of 8-month (top) and 12-month (bottom) cumulative origin and destination temperature and precipitation effects. Vertical reference lines: observed point estimates. Distributions from 100 temporally shuffled placebo panels.

population, MODIS land-use shares, MODIS net primary productivity, EDGAR sectoral $\mathrm{CO_2}$ emissions, terrain ruggedness, and national per capita GDP. Cell-level estimates are produced as shares of country GDP and converted to constant 2021 PPP US$. No climate variable enters the predictor, so the cell-level GDP series is exogenous to the within-cell temperature variation that identifies our regression. The series covers 2012–2022. We use this full BFI window rather than restricting to the 2019–2022 bilateral-flow window because the cell and country-year fixed effects identify β from within-cell year-to-year temperature variation; truncating to four years would sharply reduce that variation and the precision of β without changing what is identified. We pair the cells with CRU TS v4.09 (Harris et al. 2020) monthly temperature and precipitation aggregated to annual means on the same 0.5° grid. Restricting to cells in developing-country origins and dropping cells with the synthetic floor-coded value the BFI allocator uses for non-positive interpolations yields a panel of roughly 460,000 cell-years across the 127 origin countries.

The grid-level income regression is

$$\ln \text{GDPpc}_{c,t} \;=\; \beta\, \mathrm{TMP}_{c,t} \;+\; \gamma\, \mathrm{PRE}_{c,t} \;+\; \alpha_c \;+\; \alpha_{i(c),t} \;+\; \varepsilon_{c,t}, \tag{11}$$

where c indexes 0.5° grid cells, $i(c)$ is the country containing cell c, α_c is a cell fixed effect, and $\alpha_{i(c),t}$ is a country-year fixed effect that absorbs every shock common to all cells in a given country-year (national macroeconomic conditions, COVID). The coefficient β is identified from within-cell year-to-year temperature variation around the cell's climatological mean, net of country-year common shocks. Standard errors are clustered at the country level. We estimate under three within-cell weighting schemes that are the cell-level counterparts of three of the five country-level weights of Section 3.2: area-weighted (the cosine of latitude, proportional to physical cell area), population-weighted using 2015 cell population, and cropland-weighted using GAEZ+ 2015 cropland intensity.[13] Because each weight is time-invariant (area) or a fixed 2015 cross-sectional snapshot (population, cropland), it does not co-move with the within-cell year-to-year climate variation that identifies β, so it stays exogenous to the shock whether or not 2015 falls inside the estimation window. The area-weighted estimate gives the elasticity for the average unit of land; the population-weighted estimate for the cells where people live; the cropland-weighted estimate for the cells where farming takes place. The population-weighted estimate enters the decomposition (13), so that both sides load on the same underlying population.

Figure 7 reports the cell-level elasticities. The coefficient on temperature is negative in all three weighting schemes, significant at the 5 percent level for the population- and cropland-weighted estimates and at the 10 percent level for the area-weighted estimate (the figure shows

[13] These are cell-level counterparts of the dyadic weights, not literal reproductions of its aggregation formula. The fractional-area factor $A_{c,i}$ that the dyadic construction applies to each cell weight apportions a border cell between countries when cell climate is averaged up to a country-level exposure; it does not arise here, because each $0.5°$ cell is assigned whole to one country and the regression is run at the cell level rather than aggregated. Population and cropland enter as the cell's own counts, which already record how much of each is present and so require no area apportionment.

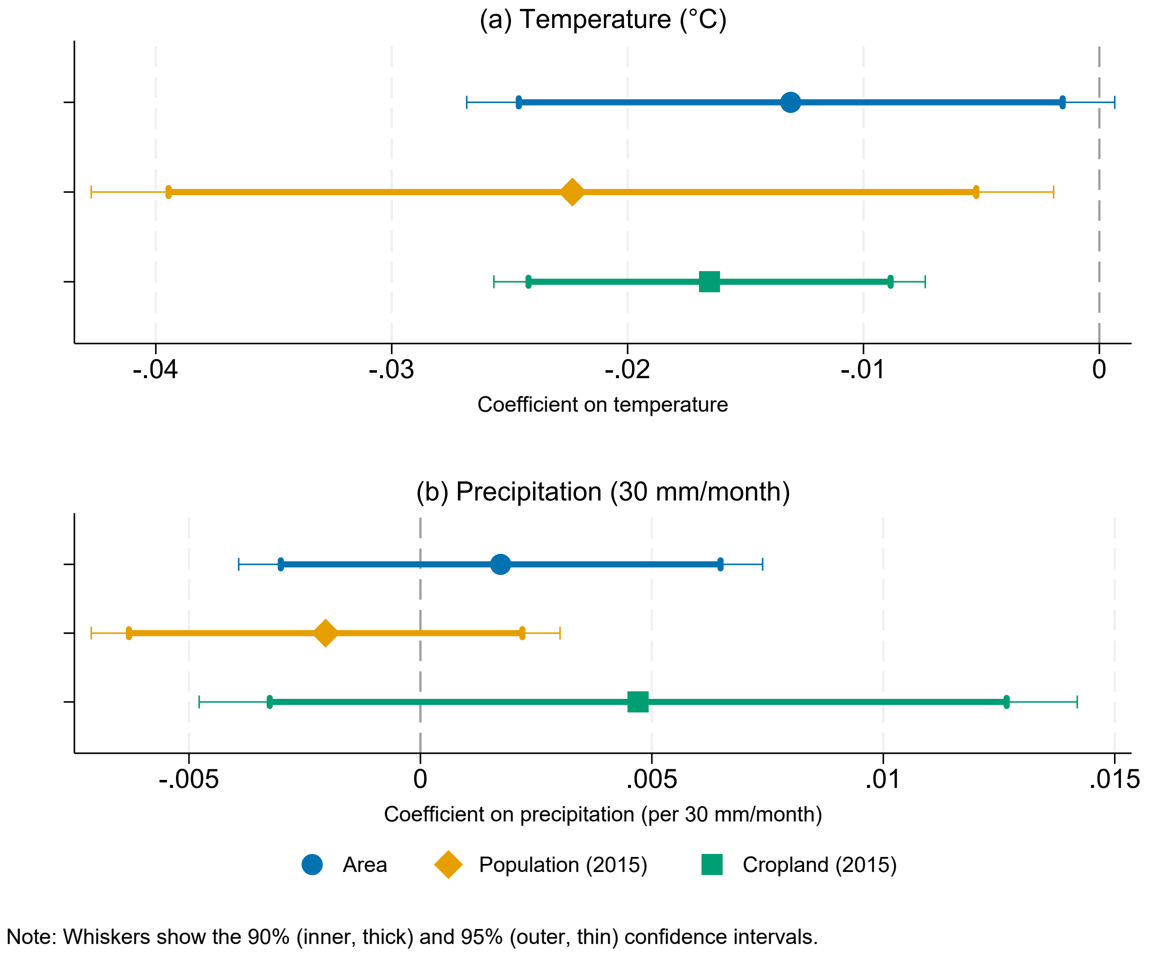


Figure 7: Cell-level climate elasticities of GDP per capita under three weighting schemes, developing-country cells, 2012–2022. Top panel: temperature (per degree Celsius); bottom panel: precipitation (per 30 mm/month of annual mean, the same rescaling as in the bilateral panel). Cell and country-year fixed effects; standard errors clustered at the country level.

both the 90 and 95 percent intervals). Read as an elasticity, a one-degree annual temperature anomaly lowers cell-level GDP per capita by about 1.3 percent for the average unit of land (area-weighted), 2.2 percent where people live (population-weighted), and 1.7 percent where cropland is concentrated (cropland-weighted): the income loss is steepest in the populated cells, the economically most relevant weight. The coefficient on precipitation is not statistically distinguishable from zero in any scheme and its sign varies, mirroring the origin-precipitation null in the bilateral panel. The income channel is therefore present and concentrated where households live; the population-weighted elasticity (-0.022) is the magnitude carried into the decomposition that follows.

5.2 Country-month cross-check from satellite nighttime lights

The BFI panel is annual; the bilateral migration panel is monthly. To test whether the activity response operates at the monthly frequency that identifies the bilateral regression, we build a country-month proxy for aggregate economic activity from satellite nighttime luminosity data. The panel covers 136 developing countries over April 2012 – December 2024 ($n = 20{,}808$ country-months). These are World Bank low- and middle-income (developing) economies with satellite coverage. As with the gridded GDP panel, we keep the full 2012–2024 window rather than restricting to the 2019–2022 flow years: the country-by-year and year-month fixed effects identify the temperature response from within-country-year monthly variation, so truncating to the flow window would cut the panel to roughly a third of its size and reduce precision without changing what is identified. The nighttime-lights panel is an independent income cross-check covering every developing country with satellite coverage, not restricted to the origins that appear in the Chi et al. (2025) flow data: it covers all 127 bilateral origins and adds nine further developing economies for which bilateral flows are too sparse to enter the migration sample. We keep the broader coverage because it makes the income test independent of the flow sample, and confirm that the result does not depend on the extra countries.[14] The proxy is the population-weighted sum of country-level radiance from NASA's Black Marble VNP46A2 daily product (Román et al. 2018), after masking for quality, snow, water, the VIIRS noise floor, and detected gas-flaring sources (see Appendix A.6). We decompose total brightness into an *intensive* margin (mean radiance per valid pixel) and an *extensive* margin (lit area in km^2), each estimated on the same panel.

The specification mirrors the bilateral migration regression:

$$\ln \mathsf{NTL}_{i,tm} = \sum_{\ell=0}^{12} \big(\beta_\ell \,\mathrm{TMP}_{i,tm-\ell} + \gamma_\ell \,\mathrm{PRE}_{i,tm-\ell}\big) + \alpha_{it} + \alpha_{tm} + \varepsilon_{i,tm}, \qquad (12)$$

where $\ln \mathsf{NTL}_{i,tm}$ is one of the three log brightness measures (total radiance, mean radiance

[14] Restricting the nighttime-lights panel to the 127 bilateral origins, dropping the nine non-origin economies, leaves the headline temperature elasticity essentially unchanged: a sustained $1\,^\circ$C anomaly lowers total brightness by 3.5 percent at eight months and 5.3 percent at twelve, the same as in the full panel.

per valid pixel, or lit area) for country i in year-month tm; the temperature and precipitation lags are grouped into the windows $\{0,\ 1\text{-}4,\ 5\text{-}8,\ 9\text{-}12\}$ of equation (10); α_{it} is a country-by-year fixed effect that absorbs national macroeconomic conditions and the COVID shock, and α_{tm} a year-month fixed effect that absorbs the common global month, so β_ℓ is identified from within-country-year monthly anomalies. Country-months are weighted by their *coverage share* (the fraction of land pixels that passed every data-quality exclusion and had at least three valid daily retrievals in the month), so observations contribute in proportion to how much of the country the satellite saw, and standard errors are clustered by country. Cumulative effects of a sustained $1\,^\circ$C temperature shock or a sustained $+30$ mm/month precipitation shock are obtained by summing the relevant lag-window coefficients.

Table 5: Climate shocks and country-month nighttime lights: cumulative effects.

	(1) $\ln$(sum radiance)	(2) $\ln$(mean rad/valid)	(3) $\ln$(lit area)
Temperature (°C)			
8m cumulative	-0.035^{***}	-0.033^{***}	-0.027^{**}
	(0.012)	(0.013)	(0.013)
12m cumulative	-0.053^{***}	-0.041^{***}	-0.042^{***}
	(0.014)	(0.015)	(0.015)
Precipitation (per 30 mm/month)			
8m cumulative	$+0.025^{**}$	$+0.033^{***}$	-0.028^{***}
	(0.012)	(0.012)	(0.010)
12m cumulative	$+0.017$	$+0.031^{*}$	-0.062^{***}
	(0.017)	(0.017)	(0.013)

Notes: OLS on the country-month nighttime-lights panel (136 developing countries, April 2012 to December 2024, $n \approx 19{,}392$ depending on the dependent variable), with observations weighted by the country-month coverage share. Column (1) is total country brightness, column (2) the intensive margin (mean radiance per valid pixel), and column (3) the extensive margin (lit area). Each climate variable enters under the grouped distributed-lag windows $\{0,\ 1\text{-}4,\ 5\text{-}8,\ 9\text{-}12\}$; cells are cumulative effects at the 8- and 12-month horizons. Country-by-year and year-month fixed effects; cluster-robust standard errors at the country level in parentheses. ${}^{*}p < 0.10$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$.

Table 5 reports the cumulative 8- and 12-month effects for the three dependent variables. The coefficient on temperature is negative and statistically significant for total country brightness at both horizons: a sustained $1\,^\circ$C anomaly lowers brightness by about 3.5% at 8 months and 5.3% at 12 months, and the intensive and extensive margins are similarly negative and significant. The activity response works through both margins on a timing that matches the bilateral regression. The precipitation response is margin-dependent: a sustained $+30$ mm/month shock reduces lit area (the extensive margin) by about 2.8% at 8 months and 6.2% at 12 months, but raises mean radiance per valid pixel and, at the 8-month horizon, total brightness. Precipitation therefore has no single-signed effect on aggregate activity, whereas the temperature response is uniformly negative; on temperature the NTL panel returns the sign of the cell-year BFI panel at the monthly frequency on which the bilateral regression identifies. The mixed-sign precipitation response is itself informative: with no consistent effect on aggregate origin activity, precipitation offers no sign-coherent income channel through which to move migration. This matches the migration evidence, where origin precipitation is insignificant in both the cell-level grid (Figure 7) and the bilateral panel. The income mechanism thus predicts the origin-precipitation null, not only the origin-temperature effect.

5.3 Implied income-migration elasticity

If the climate shock operates mainly through the income channel ($\varepsilon_{A,T} \approx 0$), the total bilateral elasticity factors is

$$\varepsilon_{m,T} = \frac{d \ln m}{d\delta} = \underbrace{\frac{\partial \ln m}{\partial \ln y}}_{\varepsilon_{m,y}} \cdot \underbrace{\frac{d \ln y}{d\delta}}_{\varepsilon_{y,T}} = \varepsilon_{y,T} \times \varepsilon_{m,y}, \tag{13}$$

which identifies $\varepsilon_{m,y}$ as the ratio of the bilateral and income elasticities on the same origins, without instrumenting income with climate. The BFI panel is annual, so the apples-to-apples horizon for the bilateral coefficient is the 12-month cumulative effect. A one-degree temperature anomaly reduces population-weighted cell-level $\ln$GDP per capita by 0.022 log-points (annual) and reduces bilateral flow by 0.057 log-points at 12 months. The ratio is

$$\varepsilon_{m,y} = \frac{-0.057}{-0.022} \approx +2.6.$$

Each one-percent climate-driven decline in origin GDP per capita is associated with roughly a 2.6 percent decline in bilateral migration flow. Using the cropland-weighted income estimate yields $\varepsilon_{m,y} \approx +3.5$; the area-weighted income estimate yields $\varepsilon_{m,y} \approx +4.4$ (all ratios computed from unrounded coefficients). All three are positive and large, consistent with a developing-origin sample in which the migration-cost constraint binds. The factorisation assumes the shock works through income; if the amenity channel is also operative ($\varepsilon_{A,T} < 0$), it raises outflows and partly offsets the income-driven contraction, so the recovered $\varepsilon_{m,y}$ is a lower bound on the income-only elasticity.[15]

[15] Estimated symmetrically, with a lagged dependent variable on both the bilateral-flow and the cell-income sides, the implied elasticity is $\varepsilon_{m,y} \approx 3.0$, close to the value above: flow and income are comparably persistent, so the short-run versus long-run rescaling largely cancels in the ratio. We also exploited the bilateral structure of the data by interacting origin temperature with corridor characteristics. Contiguity, common language, common religion, colonial ties, and distance show no statistically significant interaction. The exception is destination income: the origin-temperature elasticity is significantly more negative for corridors to richer destinations. This is what the financial-constraint mechanism predicts, since reaching a high-income destination is more expensive (visas, travel, and the financial reserves richer-country immigration regimes require), so the budget constraint binds harder and a climate-driven income loss reduces the flow by more.

5.4 A non-climate origin shock

The constraint mechanism is not specific to climate: any destructive origin shock that erodes income and the capacity to finance a move should contract cross-border migration. We test this with the calibrated monthly conflict-risk indicator of Aizenman et al. (2026), built from GDELT event data and scaled to $[0, 1]$, available at the country-month level (internal conflict risk) and the directed country-pair-month level (interstate risk) over 2015–2026.[16] We enter the indicator in the bilateral regression under the same grouped distributed-lag structure as the climate variables: windows $\{0,\ 1\text{-}4,\ 5\text{-}8,\ 9\text{-}12\}$, the four-way fixed-effect structure, and dyad-clustered standard errors.

Table 6 reports the cumulative grouped-lag effects of the three conflict-risk indicators on bilateral migration, with and without the climate grouped-lags. The coefficient on origin internal conflict risk is negative and statistically significant: a cumulative -0.44 at the 8-month horizon and -0.83 at 12 months in the specifications with only conflict indicators, and -0.34 and -0.70 once the climate grouped-lags are added. Both horizons are statistically significant. Because the indicator runs on $[0, 1]$, the twelve-month coefficient implies that moving an origin from median internal-conflict risk to the top percentile contracts bilateral migration flows by roughly one third. A non-climate origin shock, raising political tensions, thus reproduces the negative bilateral elasticity. The headline climate coefficients are unchanged by the inclusion of conflict risk (origin temperature -0.060 at eight months, destination temperature -0.058, origin precipitation insignificant, destination precipitation -0.062): the climate response is not proxying for conflict.

A binding-constraint reading of these results carries a further implication: this non-climate shock should erode origin income, in the same way that a temperature anomaly does. We test this mechanism by applying the country-month nighttime-lights income specification of Section 5.2 with the origin conflict-risk index, entered as grouped distributed lags, in place of climate. A rise in origin conflict risk lowers origin brightness: the eight-month cumulative effect is negative and statistically significant, and stays negative when the climate grouped-lags are included (Table A2). Conflict thus reproduces both halves of the mechanism, depressing origin economic activity as well as contracting bilateral migration flows. Because conflict timing is less plausibly exogenous than weather, we interpret the income result as corroborative rather than as a clean causal estimate.

6 Conclusion

Climate change and the accompanying threat to livelihoods for populations endowed with only fragile resilience to economic shocks is predicted to dramatically increase the number of ‘climate refugees’ over the coming decades. In the developing countries that would likely

[16] The Global Database of Events, Language, and Tone (GDELT) is a machine-coded catalogue of politically relevant events extracted in near real time from worldwide news media. Aizenman et al. (2026) aggregate its event counts into a calibrated monthly index of conflict risk bounded in $[0, 1]$.

Table 6: Conflict risk and bilateral migration: cumulative grouped-lag effects.

	(1)	(2)	(3)	(4)
	Risk only		Risk + climate	
	8m	12m	8m	12m
Conflict-risk indicators				
Origin internal conflict	-0.444^{**}	-0.828^{***}	-0.343^{*}	-0.700^{***}
	(0.217)	(0.283)	(0.196)	(0.258)
Destination internal conflict	$+0.032$	$+0.408$	$+0.026$	$+0.398$
	(0.234)	(0.300)	(0.236)	(0.303)
Bilateral interstate conflict	-0.202	$+0.114$	-0.351	-0.059
	(0.661)	(1.036)	(0.633)	(1.002)
Climate grouped-lags				
Origin temperature			-0.060^{***}	-0.052^{**}
			(0.021)	(0.024)
Origin precipitation			$+0.000$	$+0.016$
			(0.013)	(0.018)
Destination temperature			-0.058^{***}	-0.047^{**}
			(0.019)	(0.023)
Destination precipitation			-0.062^{***}	-0.085^{***}
			(0.013)	(0.020)

Notes: Poisson PML with high-dimensional fixed effects on the bilateral panel, 897,484 observations. Each conflict-risk indicator (calibrated monthly conflict risk in $[0, 1]$, GDELT-based) enters under the same grouped distributed-lag windows as the climate variables, $\{0, 1\text{-}4, 5\text{-}8, 9\text{-}12\}$; cells are cumulative effects. The risk-only columns include the three conflict indicators; the risk-plus-climate columns add the four climate grouped-lags. A non-climate origin shock reproduces the negative bilateral elasticity, and the headline climate coefficients are unchanged when conflict risk is included. Four-way fixed effects (origin-destination-year, origin-month, destination-month, year-month); cluster-robust standard errors at the origin-destination dyad in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

send most of the world's potential climate migrants, we find that the cross-border response to climate stress runs *opposite* to the climate-refugee narrative: a climate shock such as a high-temperature anomaly reduces international migration rather than raising it. This pattern is compatible with a resource-constrained immobility trap, and in line with the migration-cost constraint long documented for income shocks in the migration-development literature. Our results bear directly on how climate-driven migration is projected. Forecasts that multiply a global population at climate risk by a positive emigration elasticity overstate cross-border displacement for precisely the origins where the constraint binds. The populations most exposed to climate damage are typically also those least able to finance an international move, so in the near term climate stress is more likely to dampen orderly cross-border emigration from poor origins than to generate a surge of climate refugees in distant, rich countries. Whether those who stay move within their own country instead is a margin our cross-border data and analysis cannot speak to. This outcome is troubling on both economic and ethical grounds, since cross-border movement can itself be an important means of adaptation for the most vulnerable populations.

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A Online Appendix (not intended for publication)

A.1 Costly-financing extension

The model of Section 2 imposes the financing requirement (3) as a hard constraint: a household whose disposable income falls short of the bilateral cost cannot move at all. This keeps the two margins clean but rules out, by construction, the case in which a constrained household relaxes the requirement itself through informal borrowing, distress asset sales, or irregular routes. We show here that softening the constraint into a costly-financing technology nests the base model, leaves the empirical content of Section 2 unchanged, and opens a second route by which an acute amenity shock can overturn the sign of the migration response.

Replace the hard constraint with a penalty on the financing shortfall. Define the shortfall

$$s_{ij,t} \;=\; \max\bigl\{0,\; c_{ij} - (y_{i,t} - \underline{c})\bigr\}, \tag{A1}$$

the amount by which the cost c_{ij} exceeds disposable income. A household with $s_{ij,t} = 0$ can finance the move from its own resources; a household with $s_{ij,t} > 0$ must raise the difference at a cost $\Phi(s_{ij,t})$, with

$$\Phi(0) = 0, \qquad \Phi' > 0, \qquad \Phi'' > 0. \tag{A2}$$

The convexity captures the rising marginal cost of distress financing: small gaps are bridged cheaply (a short informal loan), large gaps only at steep and increasing cost (asset liquidation at fire-sale prices, high-interest debt, the risk premium on an irregular crossing). The household now migrates whenever the surplus from moving, net of both the direct cost and the financing penalty, is positive:

$$S_{ij,t} \;\equiv\; V^{\text{move}}_{ij,t} - c_{ij} - \Phi\bigl(s_{ij,t}\bigr) - V^{\text{stay}}_{i,t} \;>\; 0. \tag{A3}$$

When $s_{ij,t} = 0$ the penalty vanishes and (A3) collapses to the desirability comparison $V^{\text{move}} - c_{ij} > V^{\text{stay}}$, so unconstrained households behave exactly as in the base model.

Constrained-household response. Consider a household with $s_{ij,t} > 0$ and differentiate the surplus (A3) with respect to the climate shock $\delta_{i,t}$. A higher δ erodes income through (4), so disposable income falls and the shortfall widens at rate $\partial s_{ij,t}/\partial \delta_{i,t} = -\varepsilon_{y,T}\, y^0_{i,t} > 0$. Using $\partial V^{\text{stay}}/\partial \delta_{i,t} = u'(y_{i,t} - \underline{c})\, \varepsilon_{y,T}\, y^0_{i,t} + \varepsilon_{A,T}$,

$$\frac{\partial S_{ij,t}}{\partial \delta_{i,t}} \;=\; \underbrace{-\varepsilon_{A,T}}_{\text{amenity}} + \bigl(-\varepsilon_{y,T}\, y^0_{i,t}\bigr)\Bigl[\underbrace{u'\bigl(y_{i,t} - \underline{c}\bigr)}_{\text{income push}} - \underbrace{\Phi'\bigl(s_{ij,t}\bigr)}_{\text{financing}}\Bigr]. \tag{A4}$$

The income loss $-\varepsilon_{y,T}\, y^0_{i,t} > 0$ now pulls in two opposing directions. As before it lowers the value of staying, raising the surplus from moving (the income push, $u' > 0$). But it also widens the shortfall and, through the convex penalty, raises the marginal cost of financing the move ($\Phi' > 0$): the financing effect. The amenity term $-\varepsilon_{A,T} \geq 0$ raises the surplus regardless of the financing position. A constrained household therefore responds to the shock by migrating more iff

$$-\varepsilon_{A,T} \ + \ \left(-\varepsilon_{y,T}\, y^0_{i,t}\right)\left[u'(y_{i,t} - \underline{c}) - \Phi'(s_{ij,t})\right] \ > \ 0. \tag{A5}$$

The two limiting cases. The base model is the prohibitive-penalty limit. As $\Phi'(s_{ij,t}) \to \infty$ for any $s_{ij,t} > 0$, the financing term in (A4) dominates and the surplus from moving collapses: a constrained household never moves, whatever the income push, recovering the mechanical immobility of the financing margin (7); unconstrained households ($s_{ij,t} = 0$) are untouched and the desirability margin (6) is unchanged. The hard constraint (3) is thus the $\Phi' \to \infty$ corner of (A2), and a finite but steep Φ simply smooths the knife-edge so that a household near the threshold responds continuously rather than flipping discretely from $m = 1$ to $m = 0$.

The opposite case is distress migration. When the marginal financing cost exceeds the marginal utility of disposable income, $\Phi'(s_{ij,t}) > u'(y_{i,t} - \underline{c})$, the bracket in (A5) is negative, so the income channel on net deters the move: the household is in the trap. Yet a sufficiently large amenity loss can still satisfy (A5): if $-\varepsilon_{A,T}$ exceeds $(-\varepsilon_{y,T}\, y^0_{i,t})[\Phi'(s_{ij,t}) - u'(y_{i,t} - \underline{c})]$, a constrained household migrates despite the widening shortfall, financing the move at the convex penalty. This is the amenity-override route that the hard-constraint model cannot represent: there, an acute amenity shock acts only through unconstrained households at the desirability margin (the first bracket of (8)), whereas here it can also drive out households the base model would have classified as trapped. It offers a distress-financed route to the positive source-temperature effect on asylum applications reported by Missirian & Schlenker (2017): that observation need not arise only from unconstrained households at the desirability margin, the reading offered in the main text, but can equally reflect constrained households financing the move under an acute amenity loss, a distinction the reduced form cannot resolve.

Aggregate response and empirical content. Integrating (A4) over the corridor's households reproduces the density-weighted structure of (8) with the knife-edge financing margin replaced by a smooth band of constrained households indexed by their shortfall $s_{ij,t}$. The sign of the aggregate climate response remains an empirical question, now settled by the relative magnitude of $|\varepsilon_{A,T}|$ against the income-push-net-of-financing term integrated across that band rather than against a single threshold mass. Because the extension changes neither the reduced-form regressors nor the auxiliary implications tested in Steps 2 and 3 of Section 2 (climate must still depress origin income, and a negative non-climate origin income shock must still reproduce the negative bilateral elasticity), the empirical strategy and all reported estimates are unaffected. We retain the hard constraint in the main text for transparency and treat the costly-financing technology as the extension that accommodates an amenity-dominated, distress-financed reading where the data call for it.

A.2 Origin heterogeneity: income and baseline-temperature splits

We probe the heterogeneity of the origin-temperature elasticity along two origin characteristics. In each case the developing origins are split at the median of the characteristic, and origin temperature is interacted with an above-median dummy in the bilateral regression of Section 4; destination climate enters as ordinary controls. Figures A1 and A2 plot the cumulative 8- and 12-month origin-temperature elasticity for each half with 95 percent confidence intervals.

Figure A1 splits origins at the median of 2018 GDP per capita (PWT 11.0; Feenstra et al. 2015). The coefficient on origin temperature is negative in both halves, and the difference between them is not statistically significant at either horizon (above-median -0.080 at 8m and -0.065 at 12m; below-median -0.069 and -0.084). The elasticity therefore does not vary with origin income, and the pooled estimate is representative across the developing-origin income distribution.

Figure A2 instead splits origins at the median of baseline (climatological) temperature. The coefficient on origin temperature is not statistically distinguishable from zero in the cooler half (-0.024 at 8m, -0.011 at 12m), but is negative and statistically significant in the hotter half (-0.143 at 8m, -0.152 at 12m), and the difference between the halves is statistically significant at both horizons. The negative origin-temperature response is therefore concentrated in already-hot origins, consistent with the concave temperature–income damage function in the macro-climate literature.

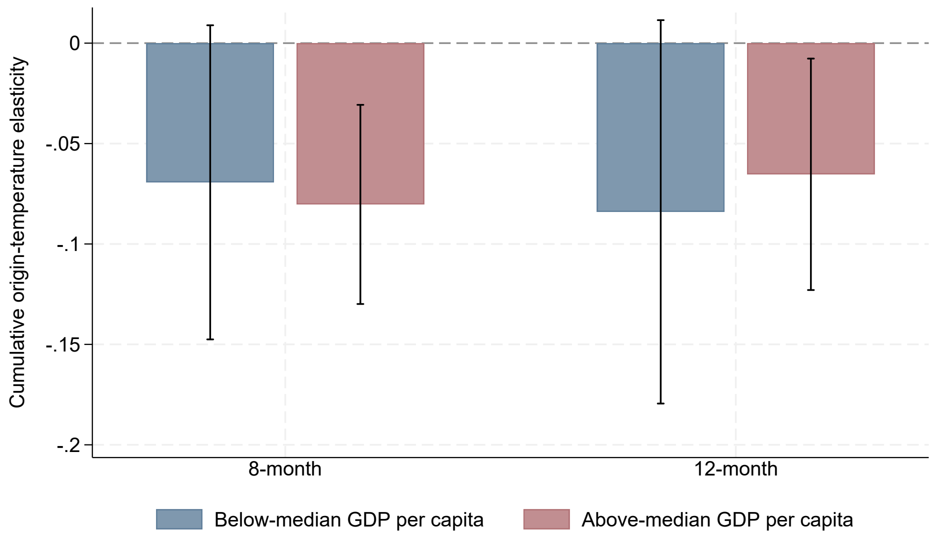

Figure A1: Origin-temperature elasticity of bilateral migration by origin income. Developing origins split at the median of 2018 GDP per capita; cumulative 8- and 12-month origin-temperature elasticity for each half, with 95 percent confidence intervals.

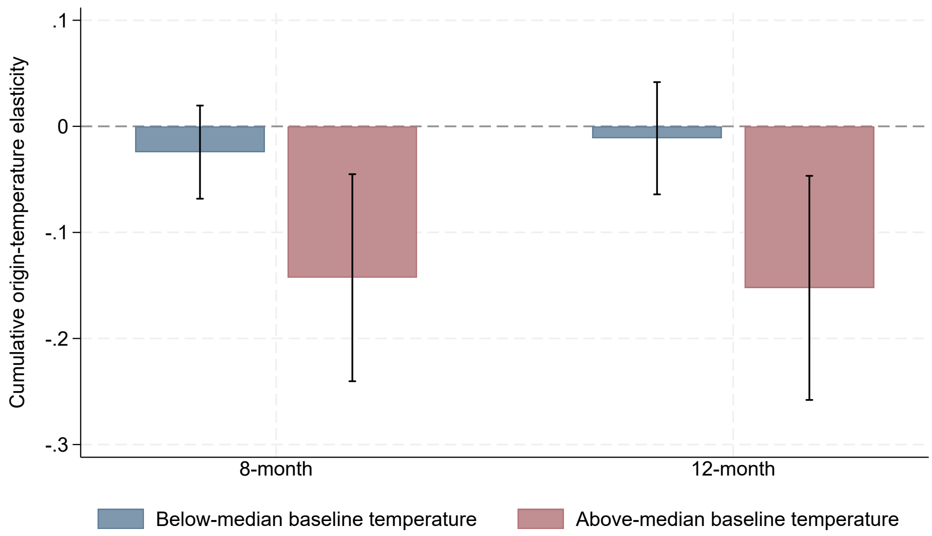

Figure A2: Origin-temperature elasticity of bilateral migration by baseline temperature. Developing origins split at the median of baseline (climatological) origin temperature; cumulative 8- and 12-month origin-temperature elasticity for each half, with 95 percent confidence intervals.

A.3 Migrant networks and the temperature response

Studies of specific corridors find that a larger pre-existing migrant network lowers the cost of moving and can turn a climate shock into a positive emigration push (Mahajan & Yang 2020, Ibáñez et al. 2026). We test this amplification channel directly by interacting the origin-temperature distributed lags with the pre-sample (2015) log bilateral migrant stock from the United Nations bilateral migrant-stock matrix (United Nations Department of Economic and Social Affairs, Population Division 2020). The stock level is time-invariant within a corridor over 2019–2022 and is absorbed by the origin-destination-year fixed effect, so only its interaction with the time-varying climate shock is identified, from the within-corridor-month variation that drives the baseline.

Table A1 reports the result for all corridors (column 1) and excluding corridors whose 2015 stock is least reliable, namely those with large reporting revisions or conflict origins (column 2). The interaction (Panel A) is negative in both columns: marginally significant in the full sample, and significant at conventional levels once the unreliable-stock corridors are dropped, where the network measure is less noisy. A larger network is thus associated with a *more* negative temperature response, not a positive push, so we find no evidence of the amplification documented for United States-bound corridors. Panel B traces the magnitude. The implied response is modest at a small diaspora (about -0.035 at the tenth percentile) and roughly doubles across the distribution, reaching about -0.068 in column 1 and -0.088 in column 2 at the ninetieth percentile, and it is statistically significant for all but the smallest diasporas. The trapping response is thus strongest exactly where the network is densest. Because the 2015 stock is correlated with standard gravity determinants such as distance, shared language, and colonial ties, the interaction is best read as describing how the response varies with corridor connectedness broadly, not as isolating the network mechanism alone.

A.4 Permutation procedure details

For each origin i, we randomly permute the 60 monthly observations of $\mathrm{TMP}_{i,\cdot}$ and $\mathrm{PRE}_{i,\cdot}$ across the months of the panel (2018–2022), breaking the alignment between i's climate at month tm and i's bilateral flow at month tm. Shuffles are independent across origins, so the cross-origin spatial correlation of weather is destroyed in the placebo. Each origin's marginal climate distribution (its climatology, and hence its mean and variance) is preserved, but the seasonal ordering and the within-origin temporal autocorrelation are not. The seed is fixed at 20260417. We repeat $R = 100$ times; the observed effect is judged causal when it lies outside the resulting placebo distribution, that is, when few or no placebo replications reproduce a coefficient as large in absolute value as the one estimated on the correctly aligned panel.

A.5 Representativeness of the 2019–2022 weather variation

A four-year window invites the concern that the 2019–2022 weather was atypical, so that the estimated elasticity might reflect a peculiar period rather than a general response. It does not. The variation that identifies the elasticity is the within-origin-month anomaly, purged of the global year-by-month path by the four-way fixed effects; the global signal of the 2020–2022 triple-dip La Niña is therefore absorbed, and identification rests on each origin's idiosyncratic deviation from its own seasonal norm. To check that this identifying variation is representative, we extend the population-weighted CRU TS country-month series (Harris et al. 2020) back to 1990 and residualise temperature and precipitation on origin-by-calendar-month and year-by-month fixed effects, exactly as in the bilateral regression. Figure A3 compares the distribution of these residual shocks over 2019–2022 against the 1990–2018 baseline. The two are statistically indistinguishable: the residual temperature shock has a standard deviation of $0.83\,^{\circ}$C over 2019–2022 against $0.82\,^{\circ}$C over 1990–2018, and a variance-ratio test does not reject equality ($p = 0.51$); the residual precipitation shock is likewise unchanged (1.53 versus 1.50 per 30 mm/month). The window thus carries ordinary weather variation, and the elasticity it identifies is not an artifact of an anomalous climatology.

A.6 Country-month NTL panel: construction

The country-month nighttime-lights proxy of Section 5.2 spans April 2012 – December 2024 for 136 developing countries ($n = 20{,}808$ country-months, of which 17 have zero land coverage in a given month and are missing by design). Monthly composites are built from NASA's Black Marble VNP46A2 daily product (Román et al. 2018) on Google Earth Engine.

For each 500 m land pixel and each calendar day we drop the retrieval if the mandatory quality flag indicates a gap-filled or borderline view angle (value 2) or no retrieval (value 255);

Table A1: Origin-temperature response and the pre-existing bilateral migrant stock.

	(1) All corridors	(2) Excl. unreliable stock data
Panel A. Coefficients		
Temperature (at median diaspora), 8-month	-0.0505^{***}	-0.0656^{***}
	(0.0191)	(0.0200)
Temperature (at median diaspora), 12-month	-0.0397^{*}	-0.0558^{**}
	(0.0217)	(0.0228)
Temperature $\times$ log migrant stock, 8-month	-0.0038^{*}	-0.0051^{**}
	(0.0023)	(0.0024)
Temperature $\times$ log migrant stock, 12-month	-0.0055	-0.0072^{**}
	(0.0034)	(0.0035)
Panel B. Implied 8-month response, by diaspora size		
10th percentile (stock of 7 bilateral migrants)	-0.0346^{*}	-0.0445^{**}
	(0.0192)	(0.0200)
25th percentile (stock of 45 bilateral migrants)	-0.0413^{**}	-0.0534^{***}
	(0.0186)	(0.0194)
50th percentile (stock of 502 bilateral migrants)	-0.0505^{***}	-0.0656^{***}
	(0.0191)	(0.0200)
75th percentile (stock of 6,542 bilateral migrants)	-0.0604^{***}	-0.0786^{***}
	(0.0213)	(0.0224)
90th percentile (stock of 44,717 bilateral migrants)	-0.0678^{***}	-0.0884^{***}
	(0.0238)	(0.0250)
Four-way fixed effects	Yes	Yes
Observations	919,532	896,300

Notes: Poisson PML estimates with high-dimensional fixed effects; the dependent variable is the bilateral migration flow. Origin and destination temperature and precipitation enter as grouped distributed lags $\{0, 1\text{-}4, 5\text{-}8, 9\text{-}12\}$; reported figures are cumulative over the 8- and 12-month horizons. Origin temperature is interacted with the 2015 log bilateral migrant stock, mean-centred across corridors, so only the stock interaction is identified (the stock level is absorbed by the origin-destination-year fixed effect). Panel A reports the temperature response evaluated at the median diaspora corridor (a 2015 bilateral stock of about 500 migrants) and the interaction slope. Panel B reports the implied cumulative 8-month response at percentiles of the bilateral stock among corridors with a positive 2015 stock; the 12-month response follows the same gradient. Column (1) uses all corridors; column (2) drops corridors whose stock is least reliable (large reporting revisions, or conflict origins). All columns include origin and destination COVID mobility controls and absorb origin-destination-year, origin-by-calendar-month, destination-by-calendar-month, and year-by-calendar-month fixed effects; standard errors clustered by origin-destination dyad in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

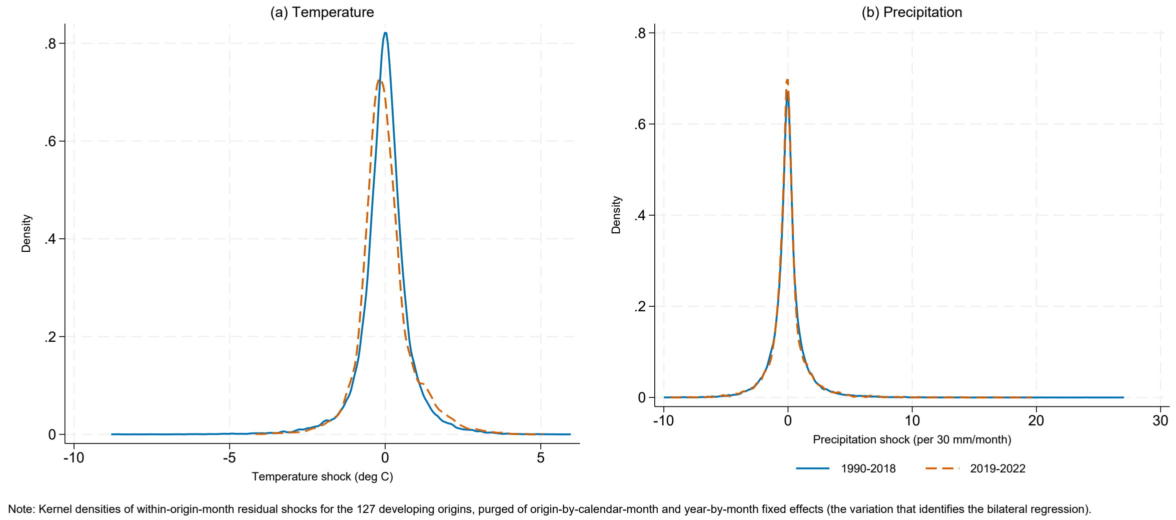

Figure A3: Distribution of the identifying weather variation, 2019–2022 versus a 1990–2018 baseline. Left panel: temperature; right panel: precipitation (per 30 mm/month). Each panel shows the kernel density of the within-origin-month residual shock for the 127 developing origins.

if the snow flag is set; if the pixel sits on a JRC Global Surface Water occurrence above 50 percent (Pekel et al. 2016); or if it falls within 1.5 km of any point in the EOG VIIRS Nightfire gas-flaring catalogue 2012–2024 (Elvidge et al. 2013, Zhizhin et al. 2021). Surviving daily radiances are clipped at the 0.3 nW/cm^2/sr noise floor. A pixel's monthly value is the mean of its surviving daily radiances when it has at least three valid daily retrievals; otherwise the pixel is missing.[1]

The country aggregate is the sum of pixel-level monthly radiances over GAUL 2015 admin-0 polygons; the *coverage share* is the fraction of country pixels surviving every mask with a non-missing monthly value. We derive three log-transformed dependent variables: total country brightness ($\ln$ sum radiance), the extensive margin ($\ln$ lit area in km^2, "lit" meaning above 1.0 nW/cm^2/sr), and the intensive margin ($\ln$ mean radiance per valid pixel). Climate exposures are the population-weighted CRU TS v4.09 (Harris et al. 2020) country-month series of the bilateral panel, with GPW v4 (Center for International Earth Science Information Network (CIESIN), Columbia University 2018) weights. The regression uses the $\{0,\ 1\text{-}4,\ 5\text{-}8,\ 9\text{-}12\}$ grouped-lag windows of the bilateral regression, country-by-year and year-month fixed effects, country-clustered standard errors, and coverage-share observation weights, so country-months in which more of the country was seen contribute more.

[1] The three-retrieval floor is permissive by design. Across the 136 countries and 153 months, the mean pixel carries 14.7 valid daily retrievals per month (median 14.6), roughly five times the floor, and only 2.6 percent of country-months fall below it. A stricter seven-retrieval floor would discard 18 percent of country-months, concentrated in the cloud-covered wet tropics (the Congo Basin, maritime South-East Asia, the Amazon fringe) and polar-night Russia: a weather-correlated sample selection in the places where climate exposure is greatest. The three-retrieval floor avoids that selection; residual variation in monthly data quality is instead absorbed smoothly through the coverage-share analytic weight. This is consistent with the minimum-observation guidance in NASA's Black Marble user guide and Román et al. (2018).

A.7 Conflict risk and origin nighttime lights

Table A2 reports the income-channel result for the non-climate shock of Section 5.4. It applies the country-month nighttime-lights specification of Appendix A.6 with the origin domestic conflict-risk index entered as grouped distributed lags: alone (column 1), against the climate lags alone (column 2), and jointly (column 3), all on a common sample. The eight-month cumulative effect of conflict risk on origin brightness is negative and statistically significant, and remains negative when the climate lags are included, confirming that the non-climate shock erodes origin economic activity at the monthly frequency. The conflict coefficient attenuates once climate is added (column 1 versus column 3), while the temperature coefficient is essentially unchanged (column 2 versus column 3), a pattern consistent with climate stress contributing to the conflict-risk signal; we do not attempt to identify that channel here. Because conflict timing is less plausibly exogenous than weather, we read the conflict result as corroborative of the income channel rather than as a clean causal effect.

Table A2: Domestic conflict risk, climate, and origin economic activity.

	(1) Conflict only	(2) Climate only	(3) Conflict + climate
Conflict risk, cumulative 8-month	-0.229^{**}		-0.175^{*}
	(0.096)		(0.095)
Conflict risk, cumulative 12-month	-0.241^{*}		-0.186
	(0.134)		(0.133)
Temperature, cumulative 8-month		-0.035^{***}	-0.036^{***}
		(0.012)	(0.012)
Temperature, cumulative 12-month		-0.053^{***}	-0.054^{***}
		(0.014)	(0.014)
Conflict-risk lags	Yes	No	Yes
Climate lags	No	Yes	Yes
Country-by-year fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Observations	19,392	19,392	19,392

Notes: Dependent variable is log total country brightness (sum of nighttime-light radiance) in the country-month panel of Appendix A.6. Conflict risk is the GDELT-based calibrated domestic conflict-risk index (in $[0, 1]$); climate is origin temperature and precipitation. All regressors enter as grouped distributed lags $\{0, 1\text{-}4, 5\text{-}8, 9\text{-}12\}$, and reported coefficients are cumulative sums over the 8- and 12-month horizons. The three columns share a common estimation sample. All regressions weight by the monthly coverage share and absorb country-by-year and year-month fixed effects; standard errors in parentheses are clustered by country. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.